University of Verona
A.Y. 2025-26

Machine Learning & Deep Learning
Deep Learning

Beyond Supervised Learning

Transfer Learning
Unsupervised Domain Adaptation

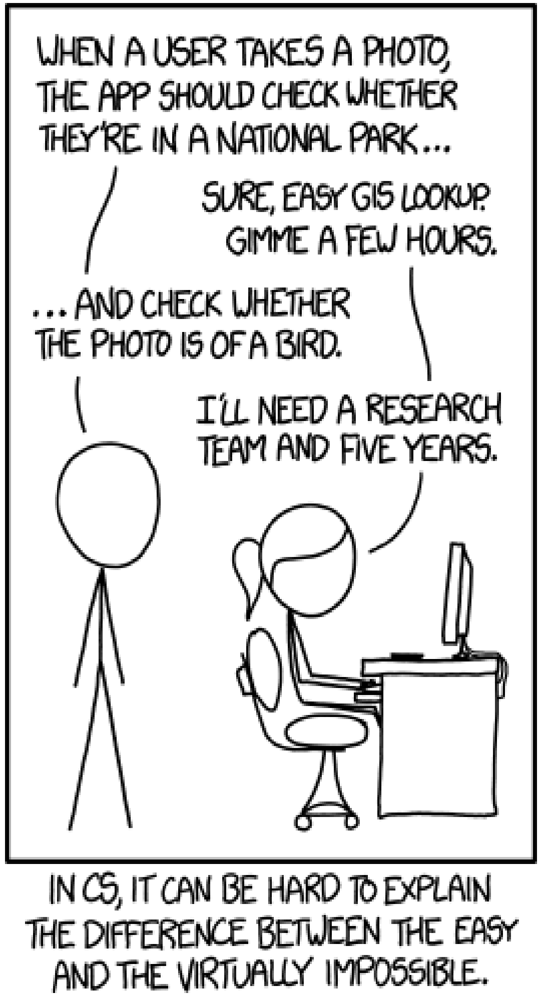
Vittorio Murino

Credits

- Tutorial by Pietro Morerio and Massimiliano Mancini
- Some slides courtesy of Prof. Elisa Ricci

Introduction

Is there a bird?



https://xkcd.com/1425/

Is there a bird?

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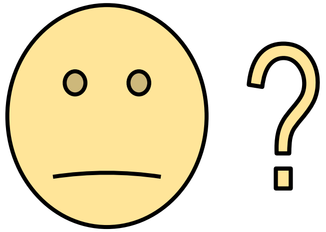
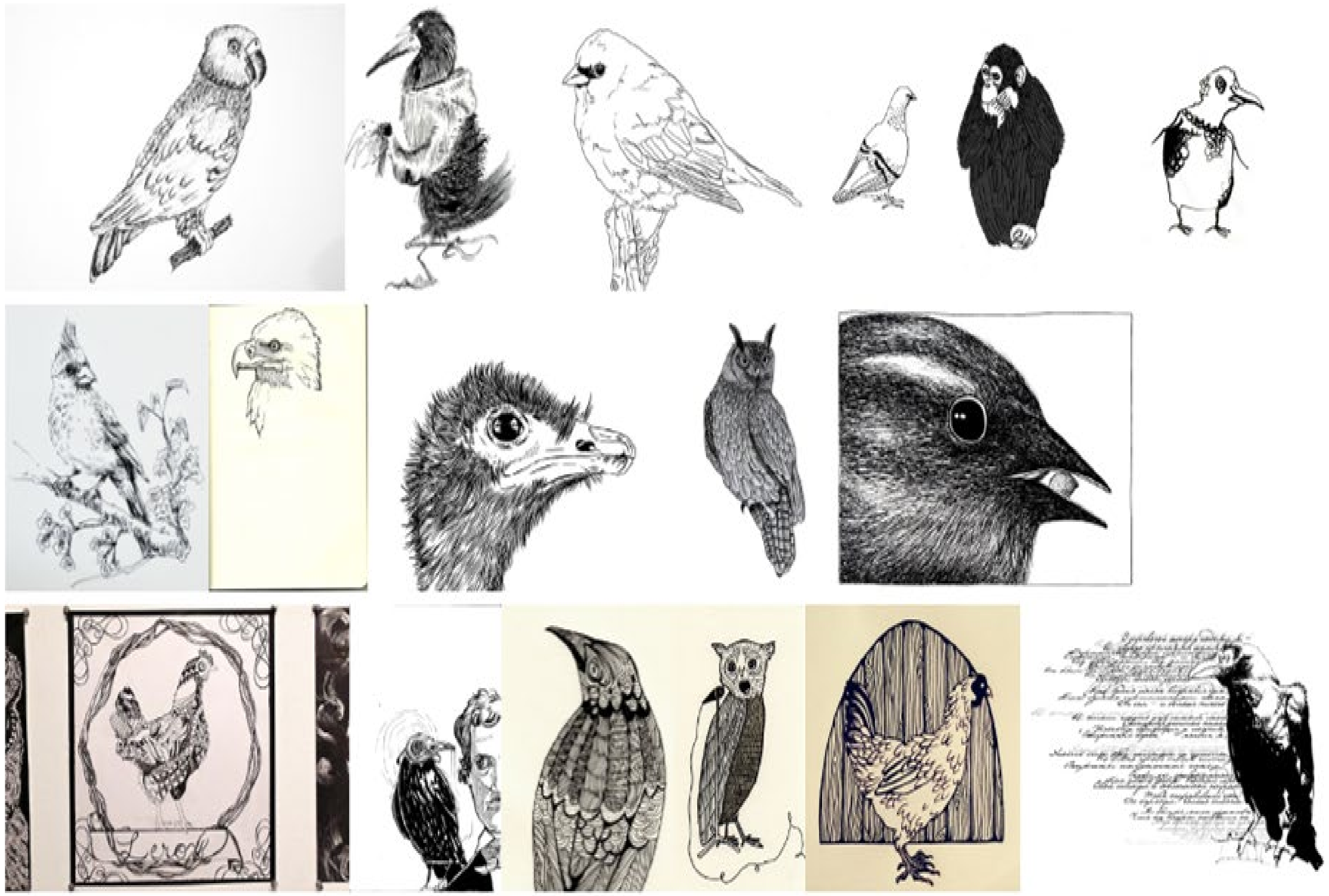
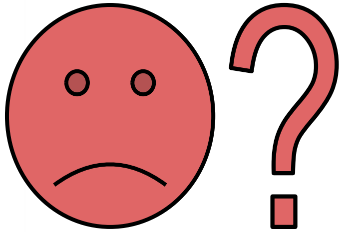


https://bam-dataset.org/

Is there a bird?

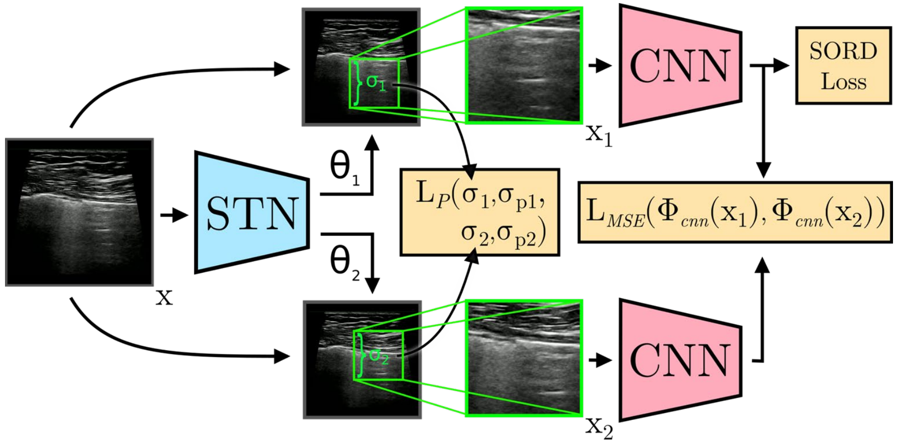


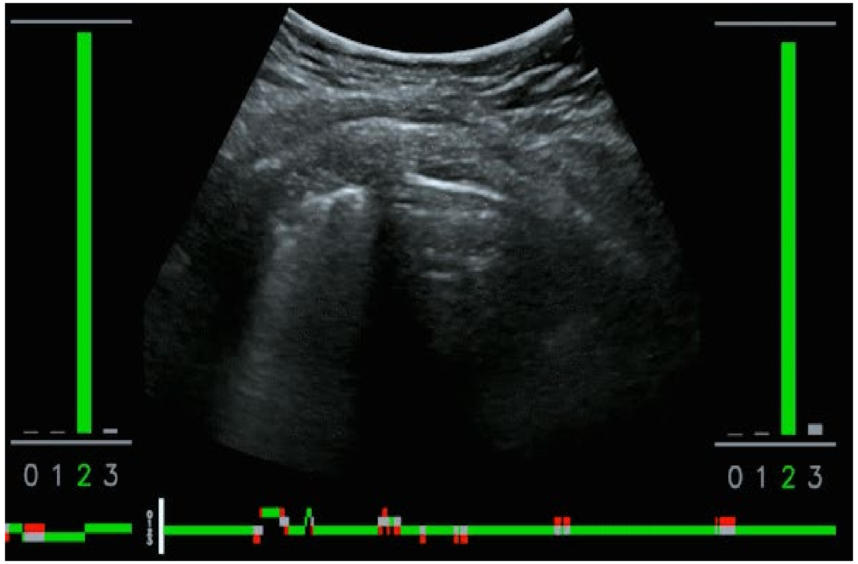


https://bam-dataset.org/

A Real-World Example

- Prediction of COVID-19 markers in lung ultrasonography images.
- Data from different hospitals (and different sensors) in Italy
- Train/test data on same hospital ≥85% accuracy.
- Performance drops of even 20% considering data from different hospitals.
- Issues in having annotated data from all the hospitals.





S. Roy et al., *Deep learning for classification and localization of COVID-19 markers in point-of-care lung ultrasound*, IEEE Trans. on Medical Imaging 2020.

Domain Shift

Visual appearance changes degrade the performance of visual recognition systems.

B. Kulis, K. Saenko, and T. Darrell, "What You Saw is Not What You Get: Domain Adaptation Using Asymmetric Kernel Transforms" CVPR, 2011.

Domain Shift

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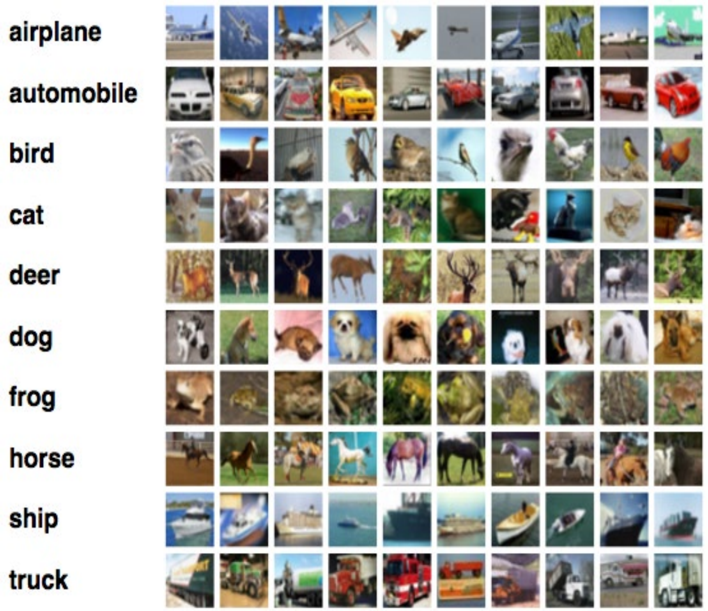

Accuracy: 85%

B. Kulis, K. Saenko, and T. Darrell, "What You Saw is Not What You Get: Domain Adaptation Using Asymmetric Kernel Transforms" CVPR, 2011.

Domain Shift

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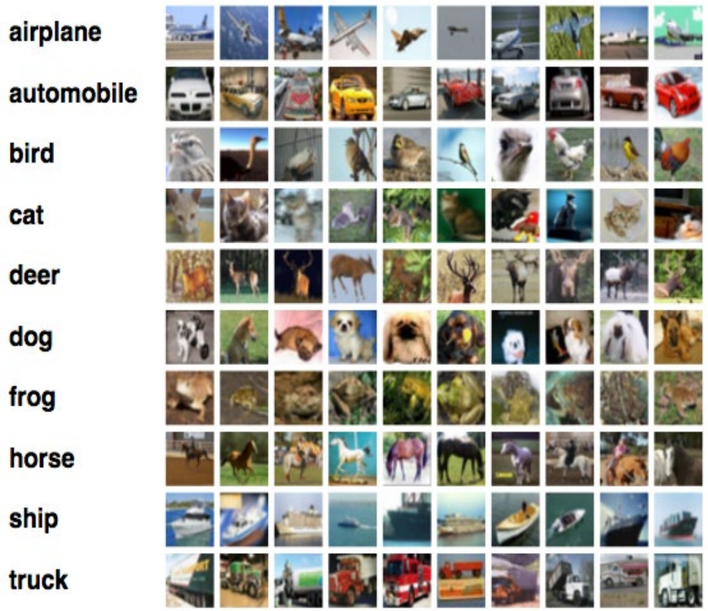

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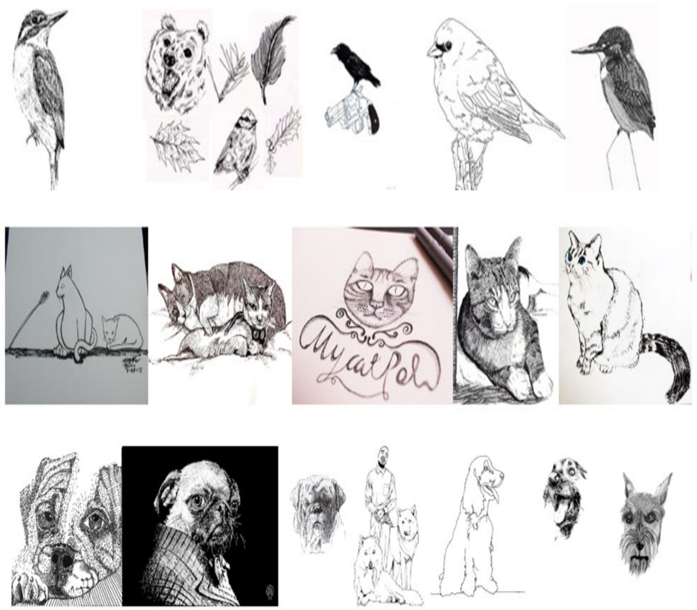


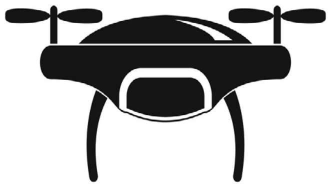
Accuracy: 55%

B. Kulis, K. Saenko, and T. Darrell, "What You Saw is Not What You Get: Domain Adaptation Using Asymmetric Kernel Transforms" CVPR, 2011.

Domain Shift: why do we care about?

Domain Shift: why do we care about?

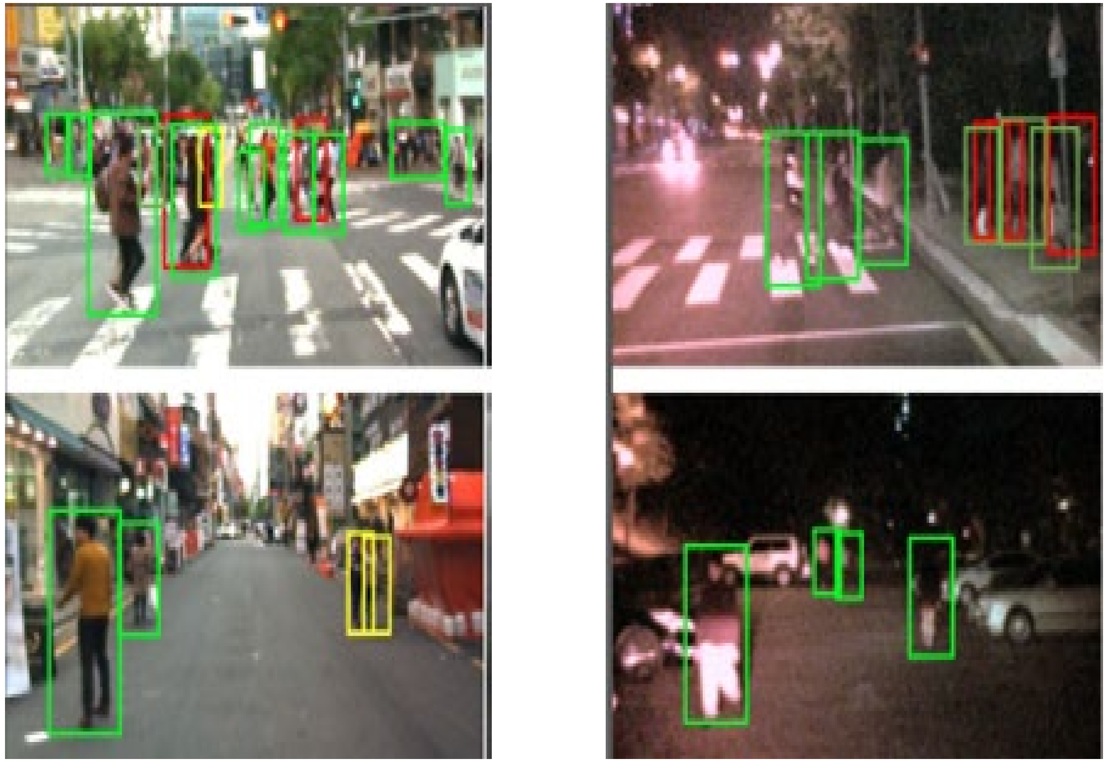
Appearance changes due to seasonal and time changes.



Z. Chen et al., "Deep Learning Features at Scale for Visual Place Recognition". ICRA 2017.

Domain Shift: why do we care about?

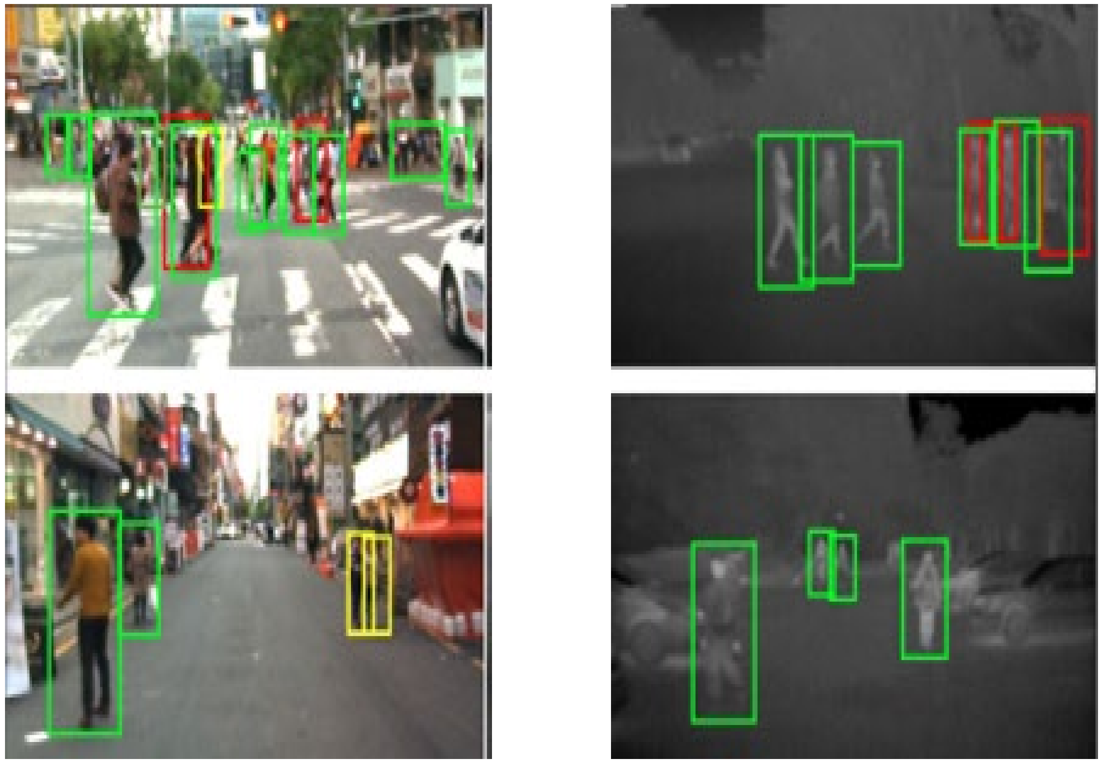
Appearance changes due to illumination conditions



S. Hwang et al., “Multispectral Pedestrian Detection: Benchmark Dataset and Baseline”, CVPR, 2015.

Domain Shift: why do we care about?

Appearance changes due to different sensors.



S. Hwang et al., "Multispectral Pedestrian Detection: Benchmark Dataset and Baseline", CVPR, 2015.

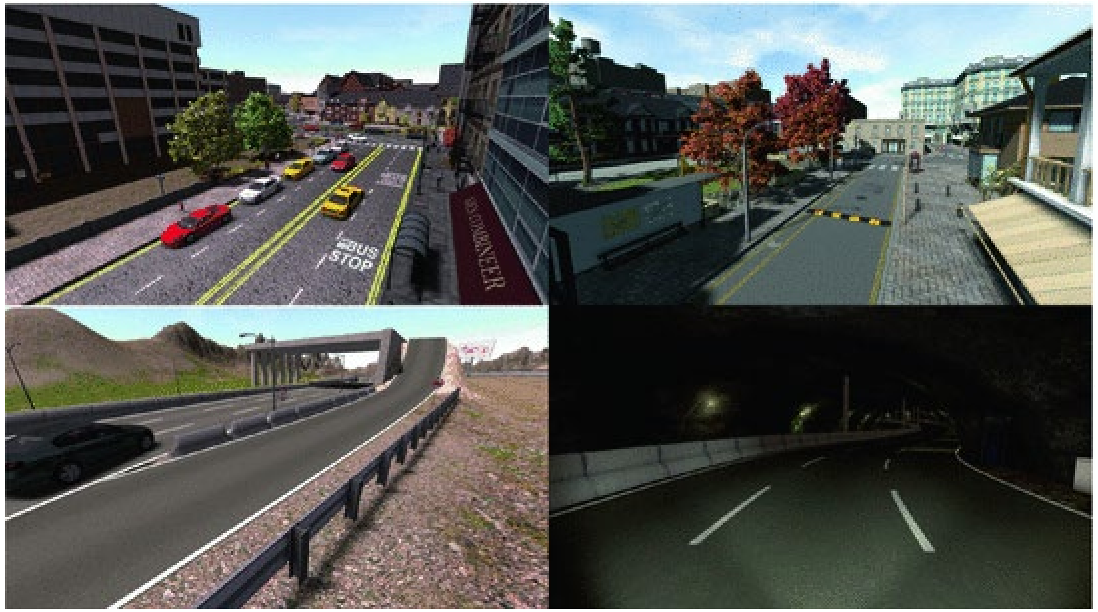
Domain Shift: why do we care about?

Overcoming costly/unfeasible data collection

https://www.zdnet.com/article/robots-to-the-rescue-searching-for-survivors-checking-on-structural-damage-in-japan/

Domain Shift: why do we care about?

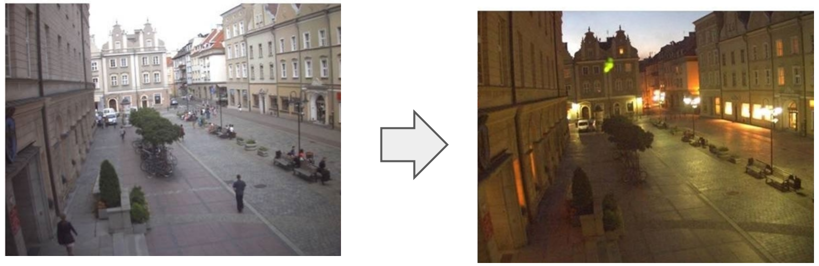
Use of synthetic data



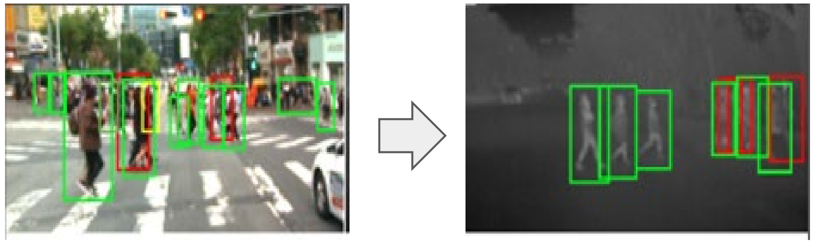
http://synthia-dataset.net/

Domain Shift is *ubiquitous* ...

Time & Environmental Changes

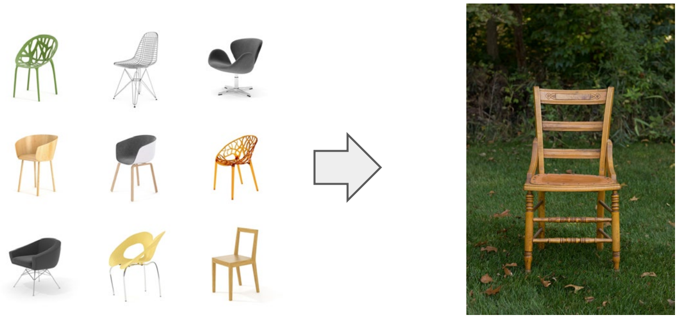


Different Modalities



Synthetic to Real Images

CAD to Real Images



Domain shift and dataset bias

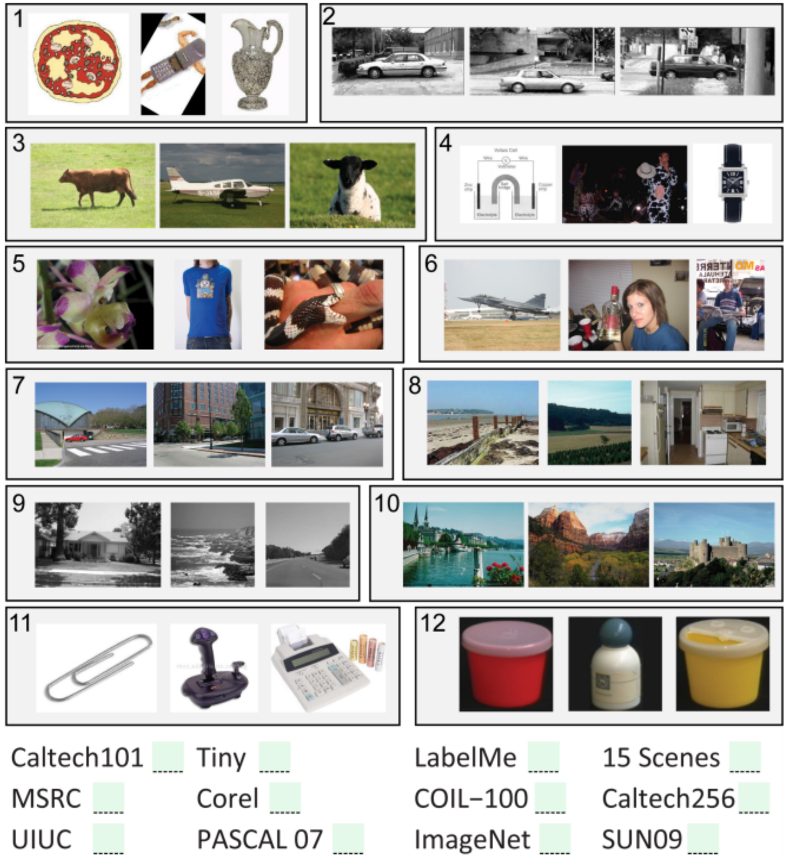
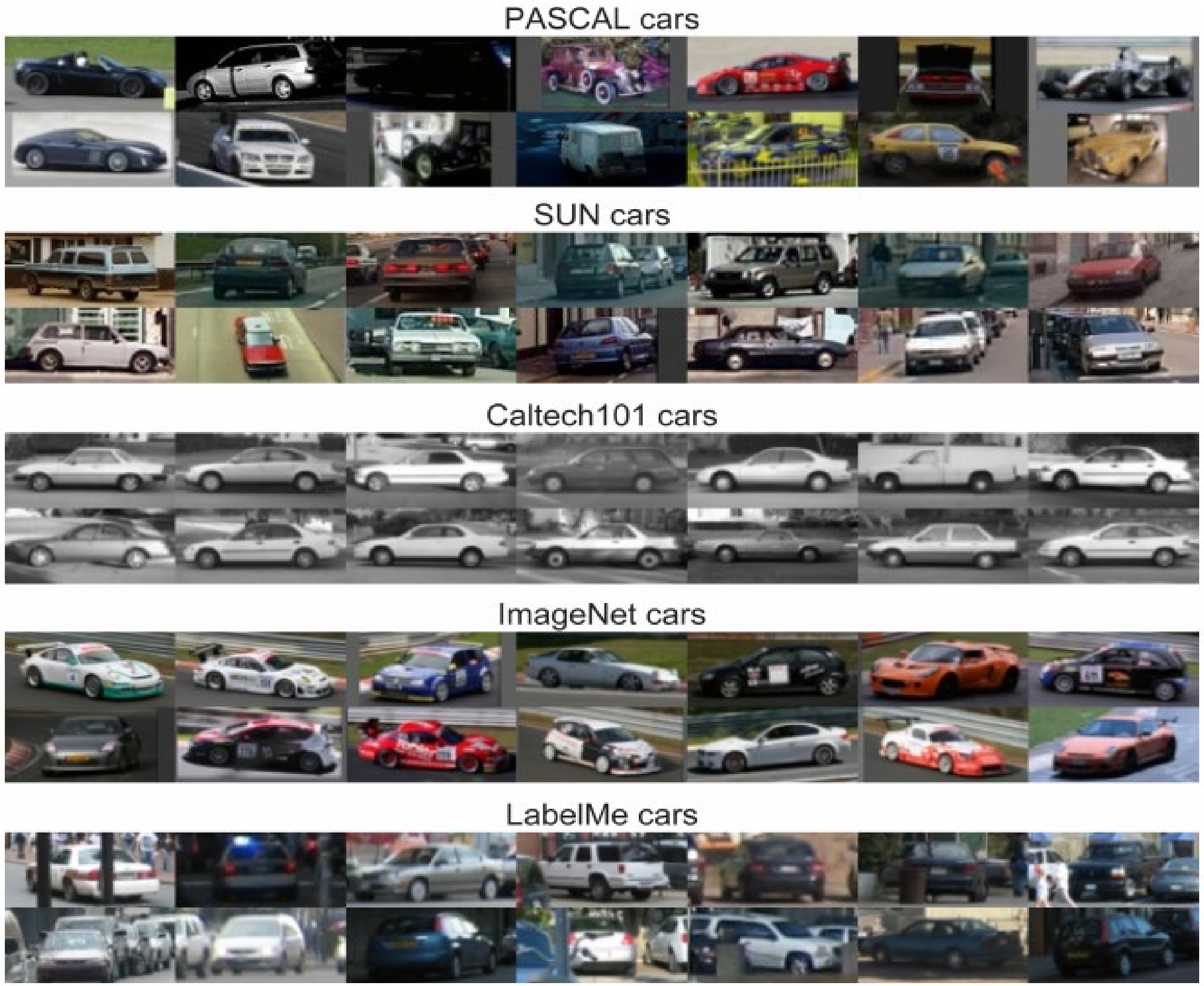


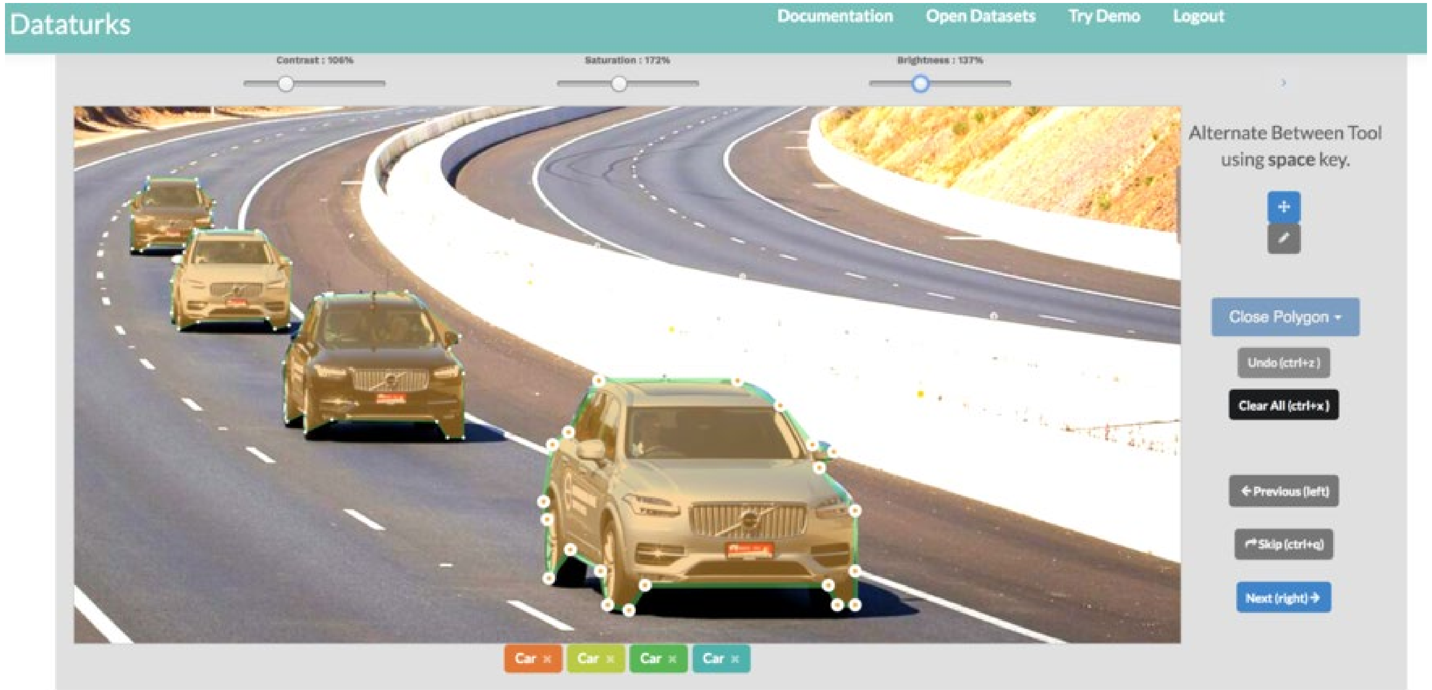
Figure 1. Name That Dataset: Given three images from twelve popular object recognition datasets, can you match the images with the dataset? (answer key below)



Torralba and Efros .,"Unbiased look at dataset bias, CVPR, 2011.

Domain Shift: how do we solve it?

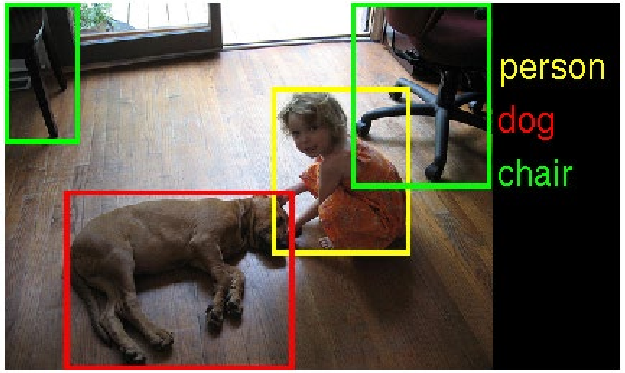
Annotate target data….



Source image

Domain Shift & Data Annotation

Data collection is costly

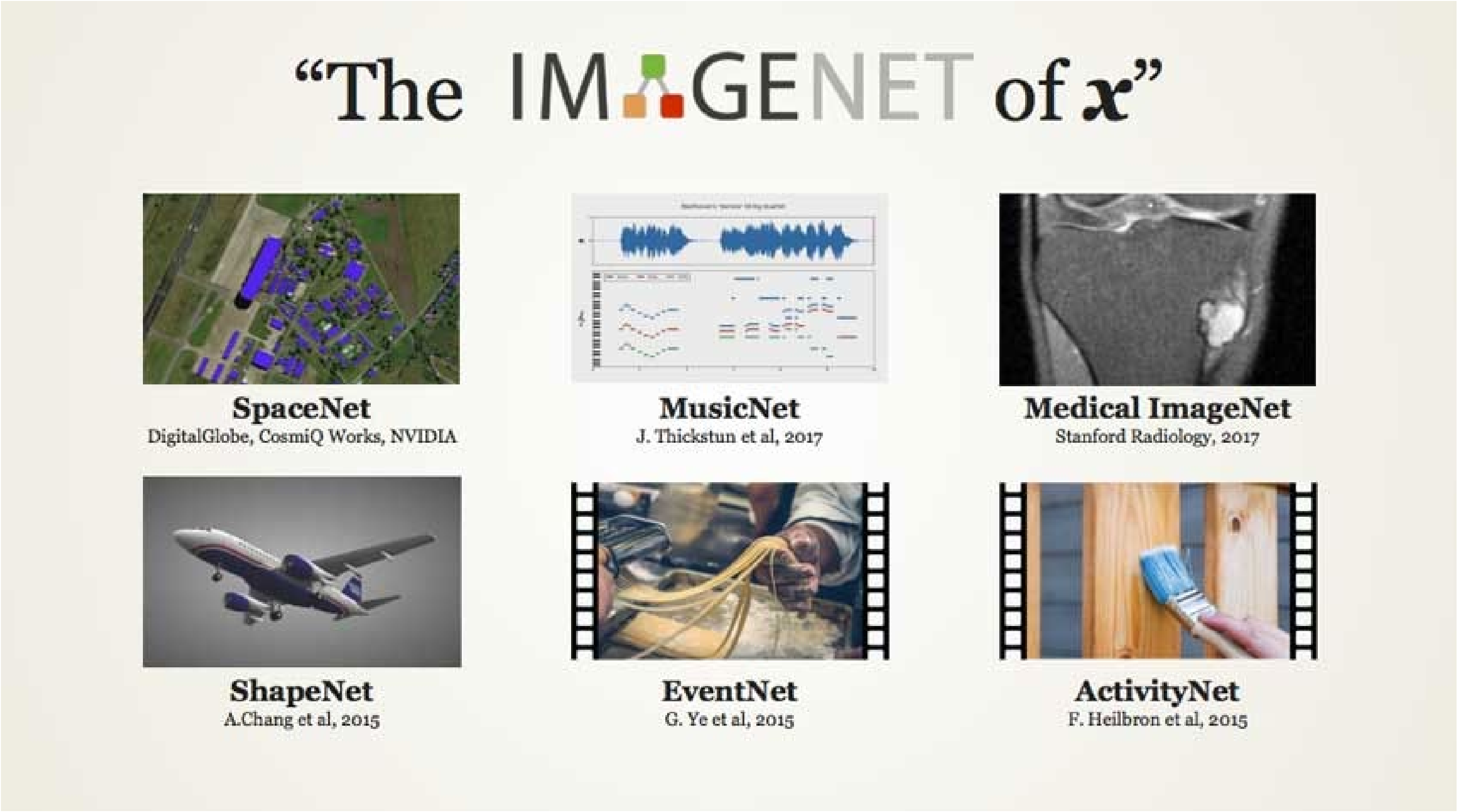


- about hand-annotated 14M images
- about 1M with bounding-boxes
- more than 20,000 categories

J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li and L. Fei-Fei, ImageNet: A Large-Scale Hierarchical Image Database. CVPR 2009.

Domain Shift & Data Annotation

We cannot annotate everything.



Slide credit: [Fei-Fei Li]

Domain Shift & Data Annotation

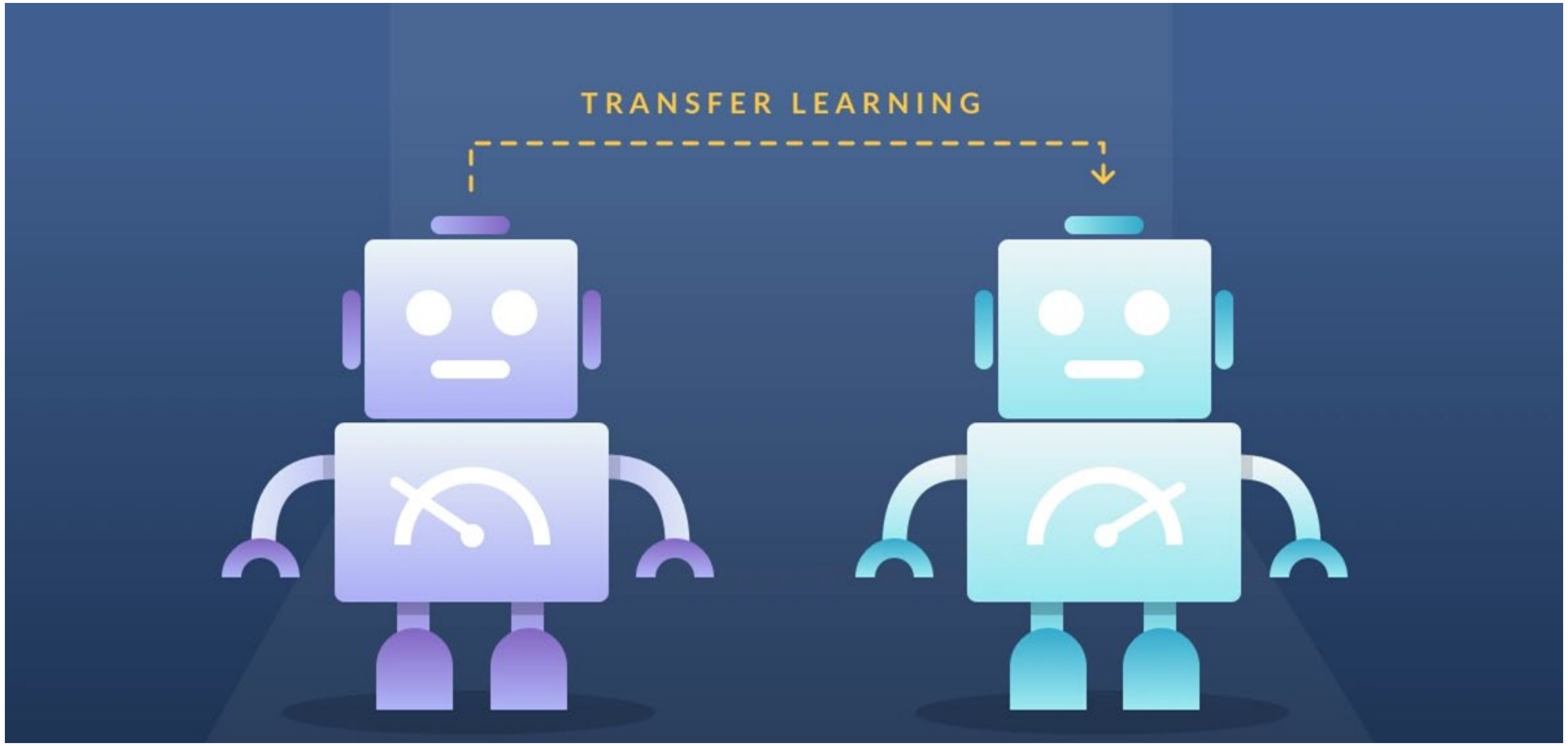
Sometimes collecting data is impossible

https://www.zdnet.com/article/robots-to-the-rescue-searching-for-survivors-checking-on-structural-damage-in-japan/

Transfer Learning

Transferring Knowledge

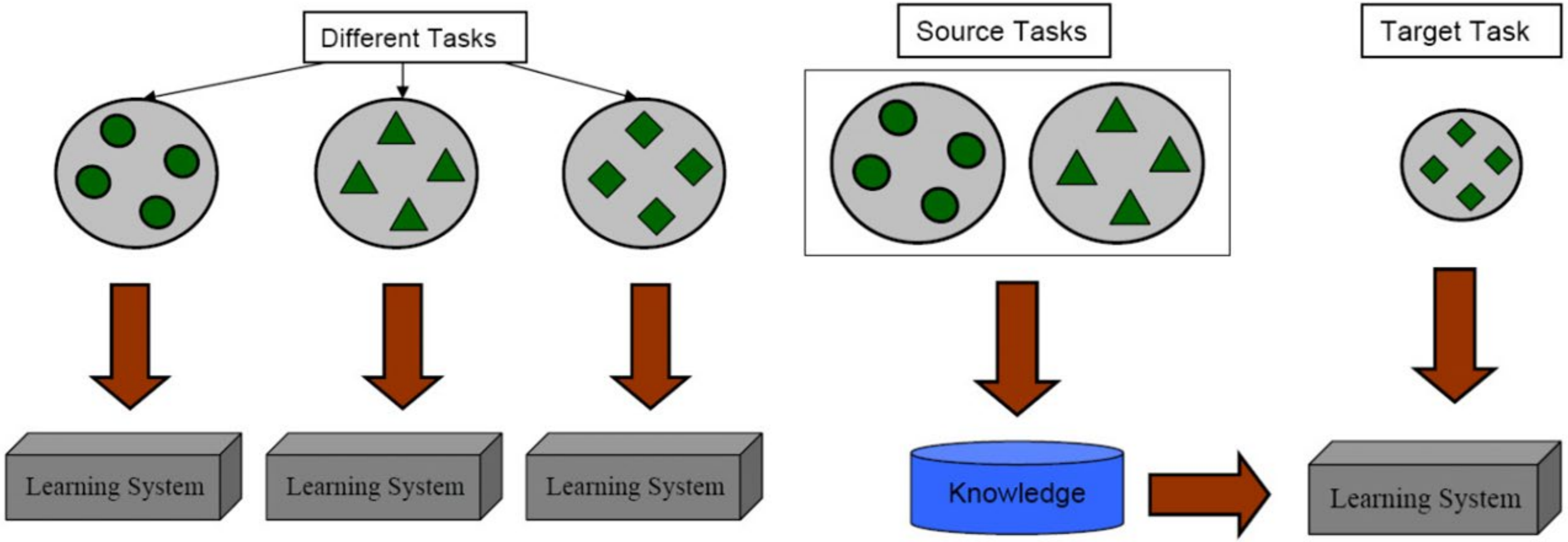
Emergence of transfer learning techniques



Source image: https://www.datacamp.com/community/tutorials/transfer-learning

Transfer Learning

Different Learning Processes between Traditional Machine Learning and Transfer Learning



(a) Traditional Machine Learning (b) Transfer Learning

Pan, S. J., & Yang, Q. (2009). A survey on transfer learning. *IEEE Transactions on knowledge and data engineering*, *22*(10), 1345-1359.

Transfer Learning: some definitions

A ***domain*** $\mathcal{D}$ *is defined as a two-element tuple consisting of:*

- *Image/Feature space* $\mathcal{X}$
- *Marginal probability P(X), where X is a sample data point.*

Given a specific domain, $\mathcal{D} = \{\mathcal{X}, P(X)\}$*, a* ***task*** $\mathcal{T}$ *consists of two components:*

- *a label space* $\mathcal{Y}$
- *an objective predictive function f(·), which is not observed but can be learned from the training data, which consist of pairs* $\{x_i, y_i\}$*, where* $x_i \in \mathcal{X}$ *and* $y_i \in \mathcal{Y}$.

Pan, S. J., & Yang, Q. (2009). A survey on transfer learning. *IEEE Transactions on knowledge and data engineering*, *22*(10), 1345-1359.

Transfer Learning: some definitions

*Given a source domain $\mathcal{D}_S$, a corresponding source task $\mathcal{T}_S$, as well as a target domain $\mathcal{D}_T$ and a target task $\mathcal{T}_T$, **the objective of transfer learning** is to learn the target conditional probability distribution $P(Y_T|X_T)$ in $\mathcal{D}_T$ with the information gained from $\mathcal{D}_S$ and $\mathcal{T}_S$, where $\mathcal{D}_S \neq \mathcal{D}_T$ or $\mathcal{T}_S \neq \mathcal{T}_T$.*

A limited number of labeled target examples, which is much smaller than the number of labeled source examples, or just unlabeled samples are assumed to be available.

Pan, S. J., & Yang, Q. (2009). A survey on transfer learning. *IEEE Transactions on knowledge and data engineering*, *22*(10), 1345-1359.

Transfer Learning: scenarios

We can consider 4 main scenarios:

- $\boxed{\mathcal{X}_S \neq \mathcal{X}_T}$ – The feature spaces of the source and target domain are different. Example: documents written in two different languages.
- $\boxed{P(X_S) \neq P(X_T)}$ – The marginal probability distributions of source and target domains are different → **Domain adaptation**
- $\boxed{\mathcal{Y}_S \neq \mathcal{Y}_T}$ – The label spaces between the two tasks are different, e.g. documents need to be assigned different labels in the target task.
- $\boxed{P(Y_S|X_S) \neq P(Y_T|X_T)}$ – The conditional probability distributions of the source and target tasks are different, e.g. source and target documents are unbalanced with regard to their classes.

Pan, S. J., & Yang, Q. (2009). A survey on transfer learning. *IEEE Transactions on knowledge and data engineering*, *22*(10), 1345-1359.

Transfer Learning: scenarios

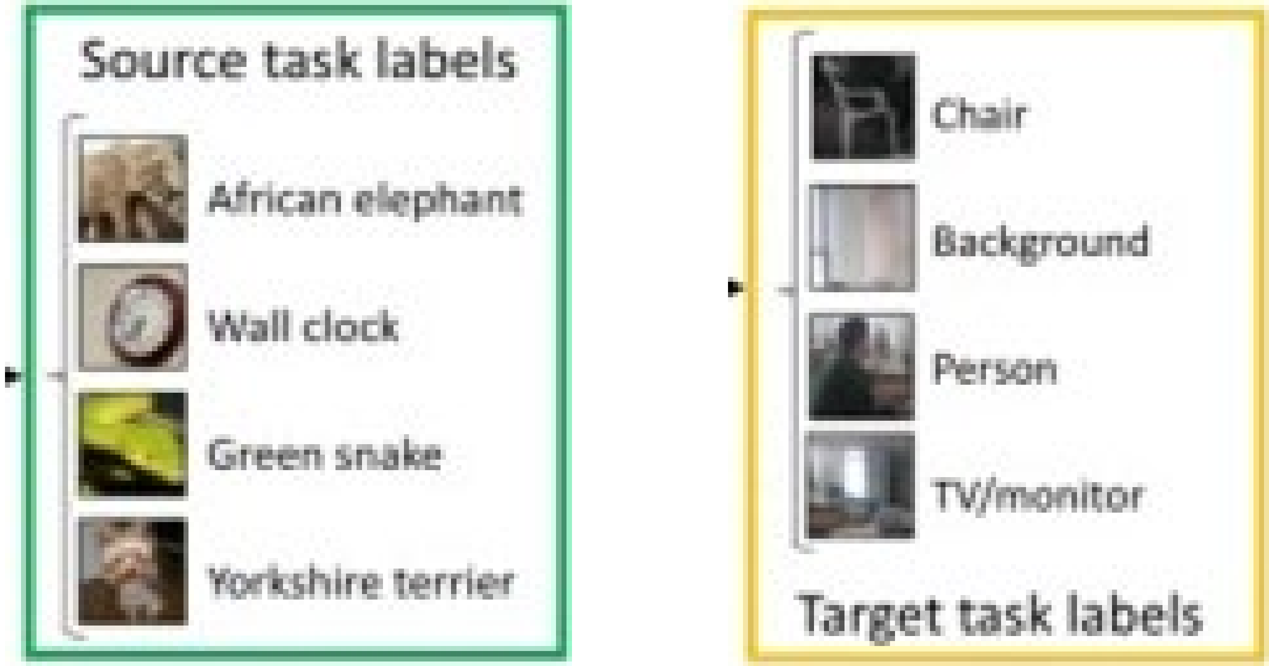
Different marginal distributions:

MNIST

SVHN

Transfer Learning: scenarios

Different label space:



Transfer Learning

Key aspects:

- **What to transfer:** Identify which part of the knowledge can be transferred from the source to the target in order to improve the performance of the target task. Understand what is domain/task-specific and what is common between the source and the target.

- **When to transfer:** We aim to improve performance on target task and not degrade them. Need to avoid *negative transfer*.

  Indeed, performance on *source tasks/domain* should be maintained

- **How to transfer:** Identify algorithmic solutions for transferring the knowledge across domains/tasks.

Domain Adaptation

A bit of notation

$\mathcal{X}$	Input space
$\mathcal{Y}$	Output space
$X \in \mathcal{X}$	Input variable (image)
$Y \in \mathcal{Y}$	Output variable (label)
$D = \{\mathcal{X}, P(X)\}$	Domain
$T = \{\mathcal{Y}, P(Y\|X)\}$	Task

F. Orabona & T. Tommasi. Tutorial on Domain Adaptation and Transfer Learning. *ECCV 2014*.

The Domain Adaptation (DA) Problem

Source Domain

$$D^s = \{\mathcal{X}^s, P(X^s)\}$$
$$T^s = \{\mathcal{Y}^s, P(Y^s|X^s)\}$$

Target Domain

$$D^t = \{\mathcal{X}^t, P(X^t)\}$$
$$T^t = \{\mathcal{Y}^t, P(Y^t|X^t)\}$$

DA problem

$$D^s \neq D^t \qquad T^s = T^t$$

F. Orabona & T. Tommasi. Tutorial on Domain Adaptation and Transfer Learning. In ECCV 2014.

The Domain Adaptation (DA) Problem

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Target Domain

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DA problem

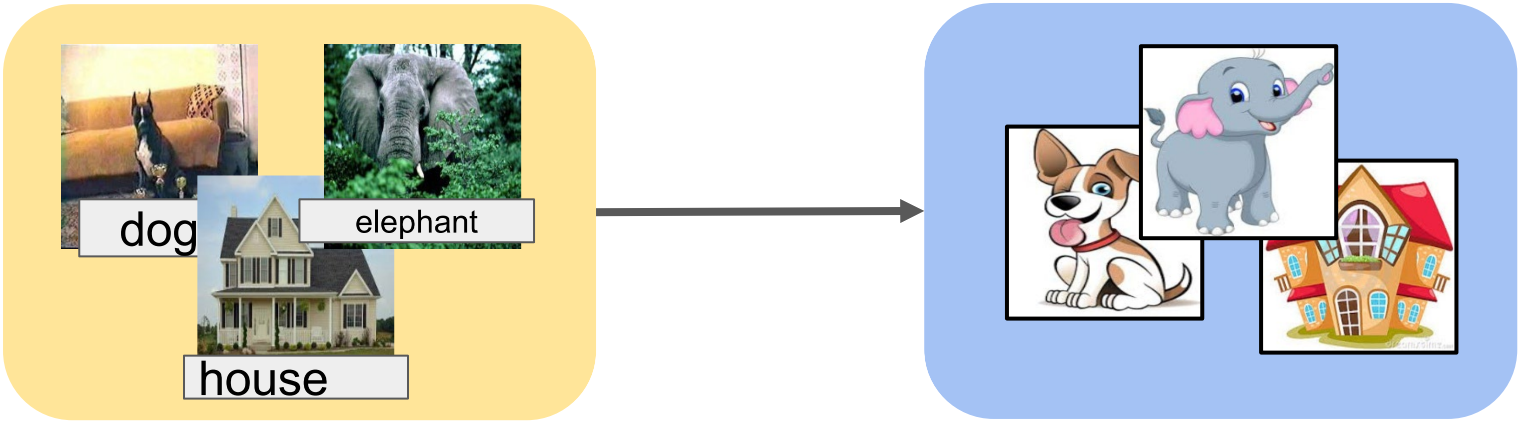
$$\mathcal{X}^s \neq \mathcal{X}^t$$
$$P(X^s) \neq P(X^t)$$

$$\mathcal{Y}^s = \mathcal{Y}^t$$

DA Landscape: unsupervised DA

Labelled
Source Domain

Unlabelled
Target Domain



$$P(X^s) \neq P(X^t)$$

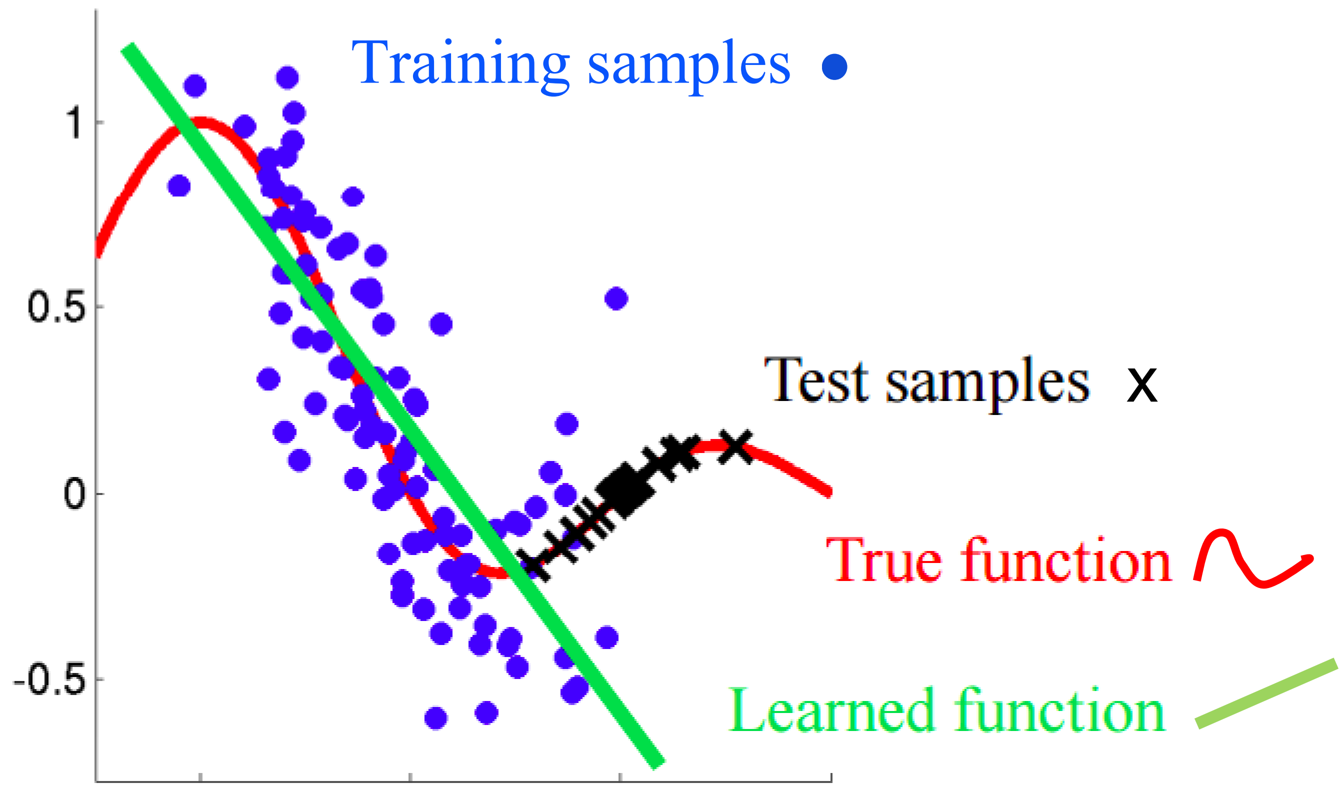
$$\mathcal{Y}^s = \mathcal{Y}^t$$

Real images vs Cartoons

{elephant, dog, house}

http://www.eecs.qmul.ac.uk/~dl307/project_iccv2017

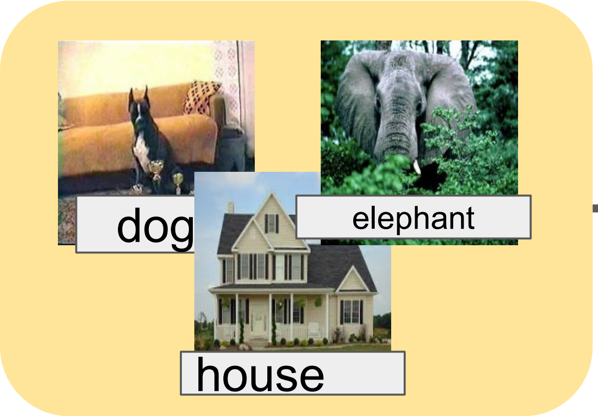
Covariate Shift Assumption



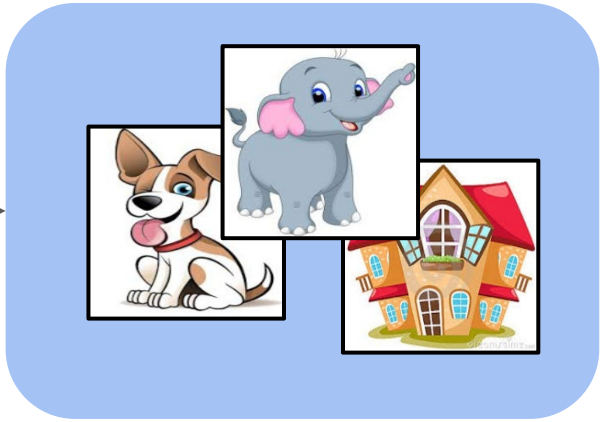
DA Scenarios

DA Scenarios: unsupervised DA

Labelled
Source Domain



Unlabelled
Target Domain



$$P(X^s) \neq P(X^t)$$

$$\mathcal{Y}^s = \mathcal{Y}^t$$

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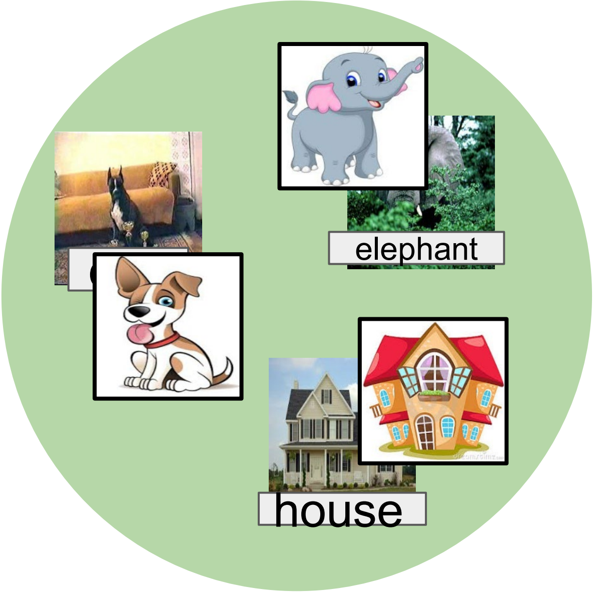
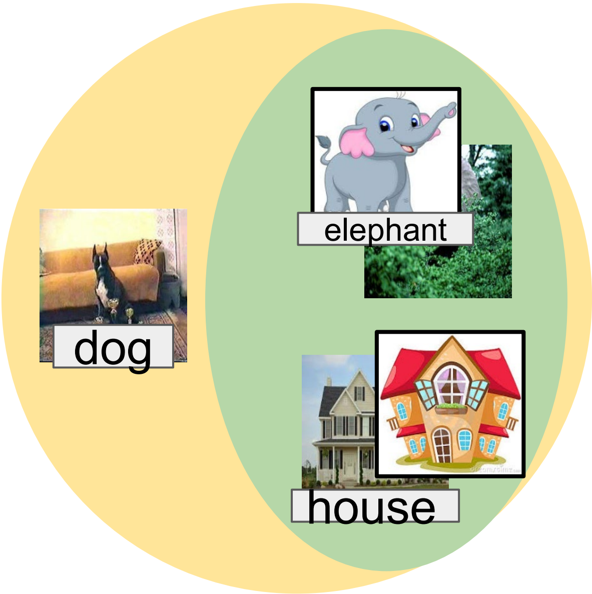
DA Scenarios: the label space

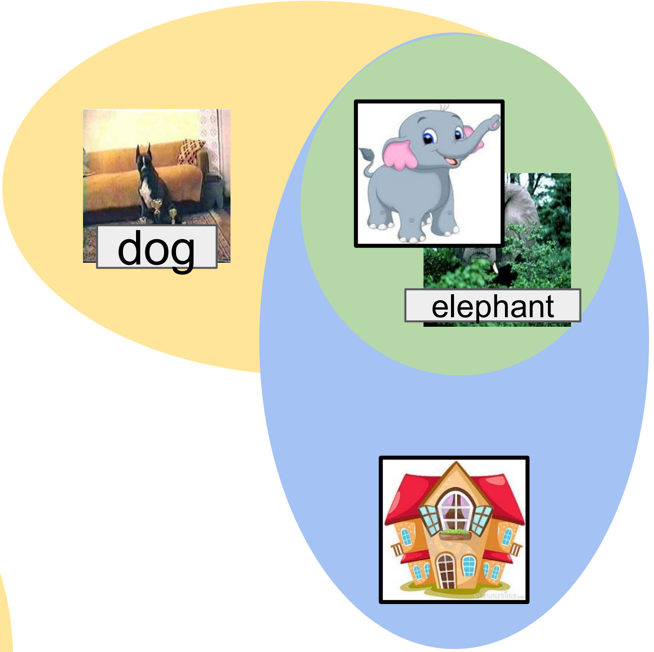
Closed Set



Partial

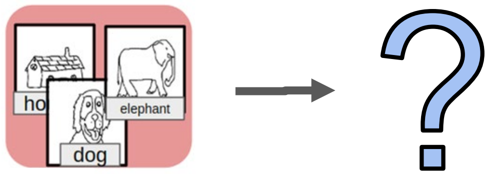


Open Set

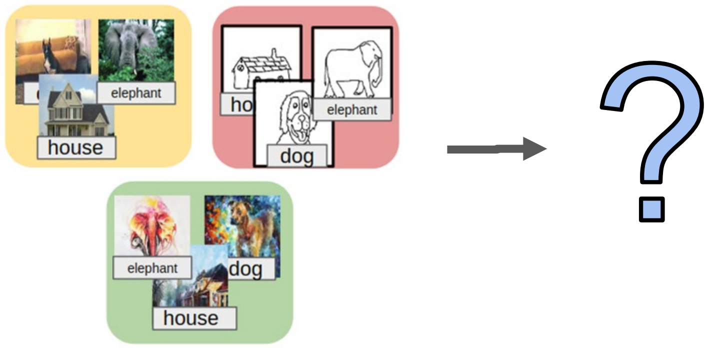


DA Scenarios: without target data!

Single-Source
Domain *Generalization*

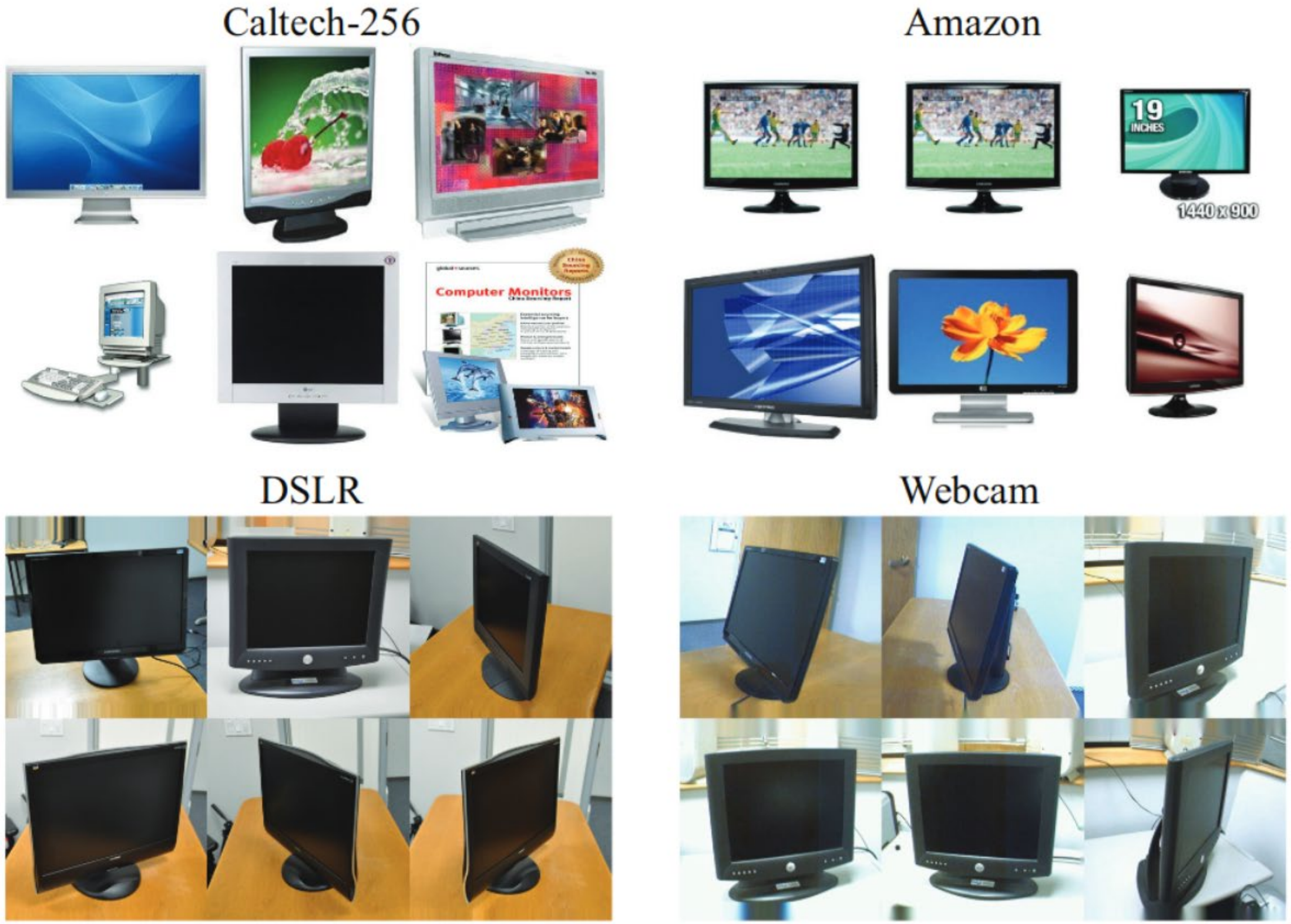


(Multi-Source)
Domain *Generalization*



A bit of history …

Benchmarks: Office31



Saenko, K., et al. "Adapting visual category models to new domains." In ECCV 2010.
Gong, B., et al. "Geodesic flow kernel for unsupervised domain adaptation." In CVPR 2012.

A bit of history

Deep Domain Adaptation

- Three main categories:
  - **Discrepancy-based:** fine-tuning the deep network with labeled or unlabeled target data to diminish the domain shift
  - **Adversarial-based:** using domain discriminators to encourage domain confusion through an adversarial objective
  - **Reconstruction-based:** using the data reconstruction as an auxiliary task to ensure feature invariance

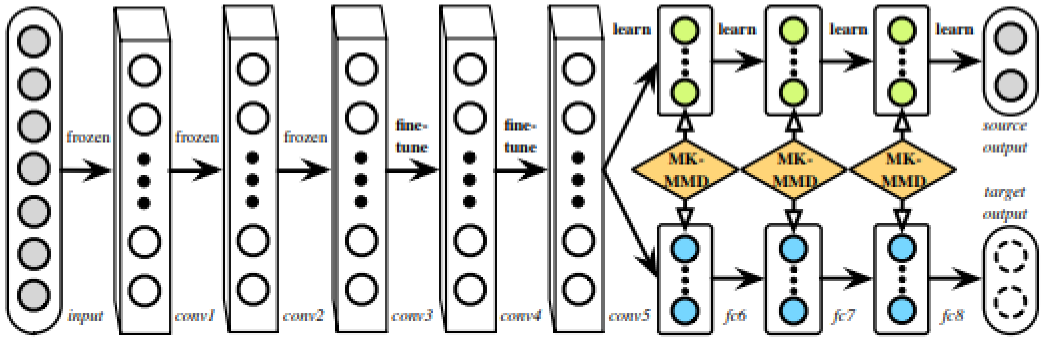
G. Csurka, "Domain adaptation for visual applications: A comprehensive survey". arXiv preprint arXiv:1702.05374, 2017

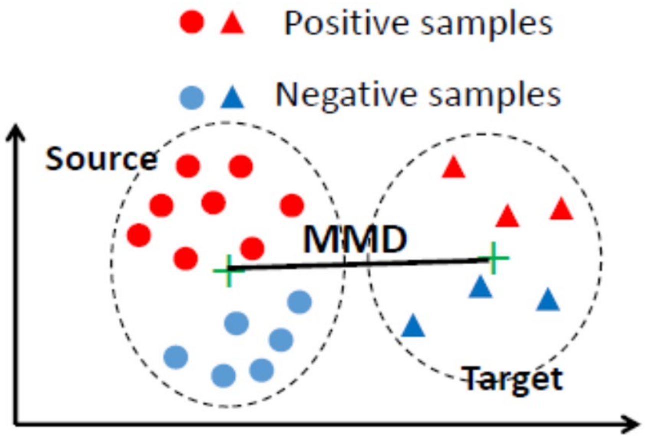
A bit of history

Deep DA – Maximum Mean Discrepancy

- Distance between embeddings of the probability distributions in a reproducing kernel Hilbert space.
- In a deep network, bottom layers capture general patterns of the data, whereas top ones are more tailored to the classification task *and thus biased towards the source/target domain*.
- **Idea**: perform a re-mapping of the feature representation in a common embedding so that such bias is removed. Such re-mapping is made at different layers and is induced by a MMD distance over a kernel function

$$\mathrm{MMD}^2 = \left\| \frac{1}{n^s} \sum_{i=1}^{n^s} \phi(x_i^s) - \frac{1}{n^t} \sum_{i=1}^{n^t} \phi(x_i^t) \right\|^2$$

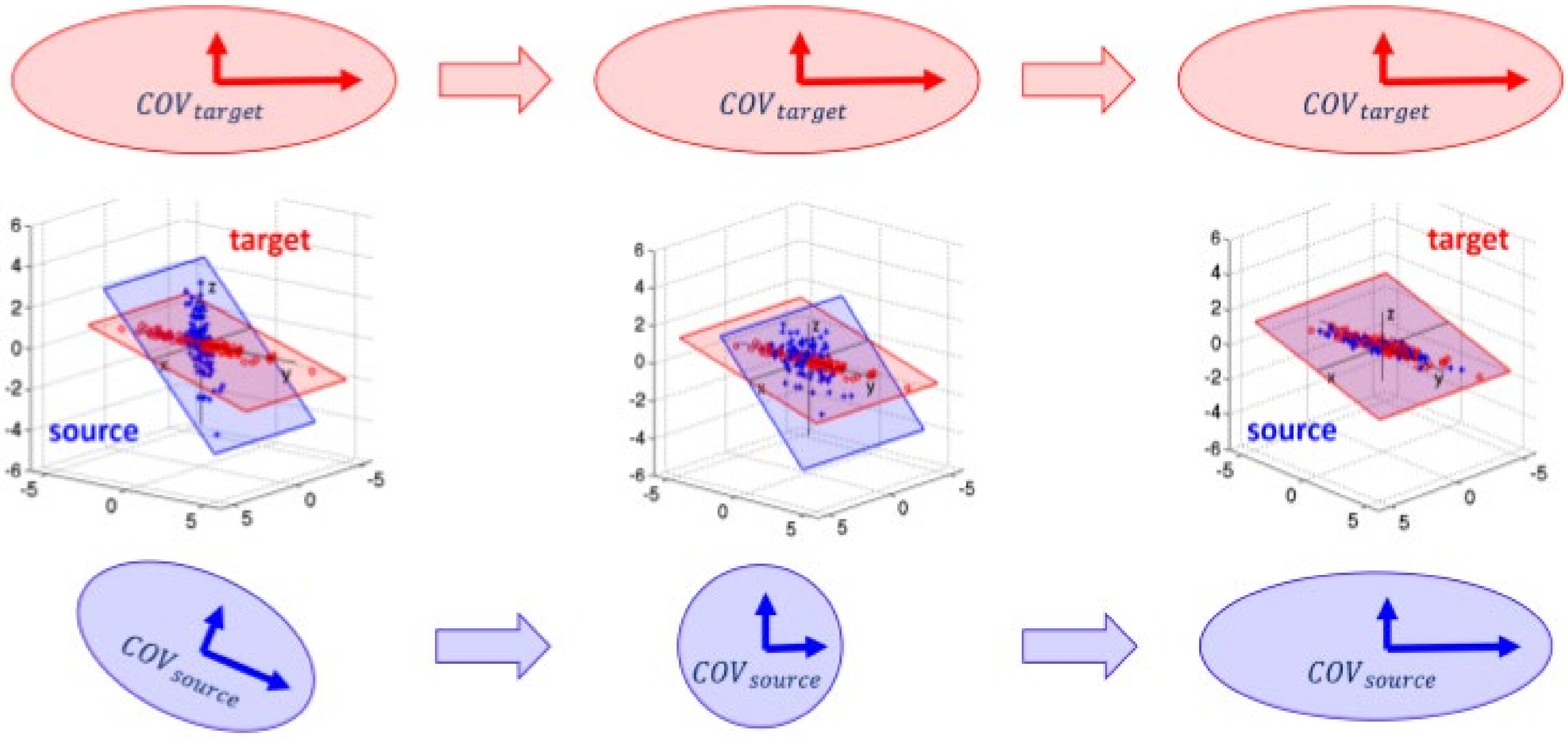




Gretton, A. et al. . "A kernel method for the two-sample-problem". In NIPS. 2007.
F. Orabona & T. Tommasi. Tutorial on Domain Adaptation and Transfer Learning. In ECCV 2014.

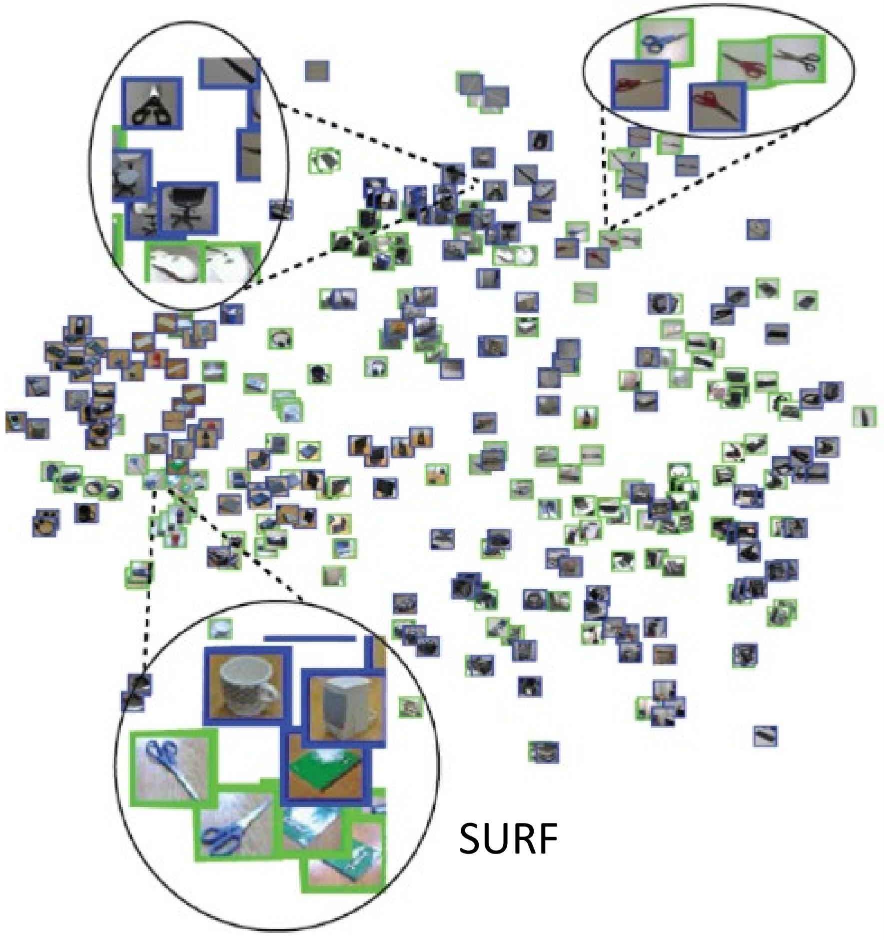
A bit of history

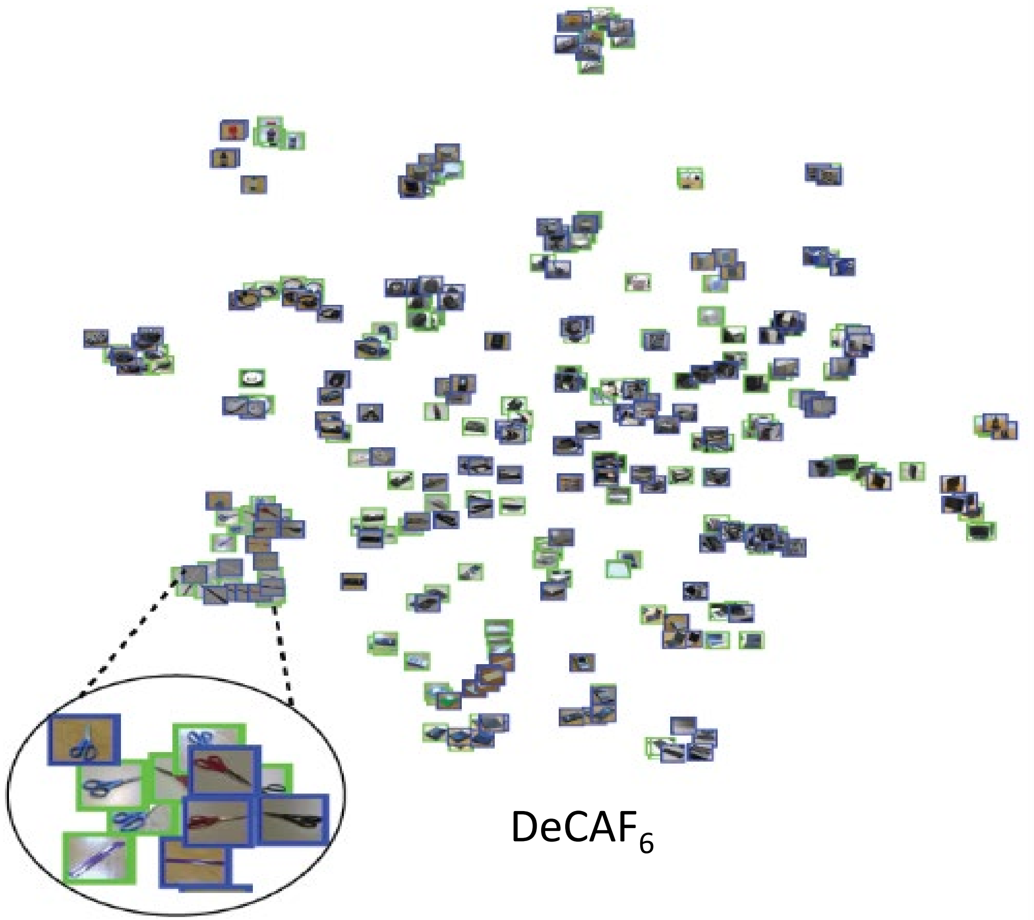
Deep DA – CORrelation ALignment, CORAL



Sun, B., et al. "Correlation alignment for unsupervised domain adaptation". In Domain Adaptation in Computer Vision Applications, 2017.

CNN & Universal Representations





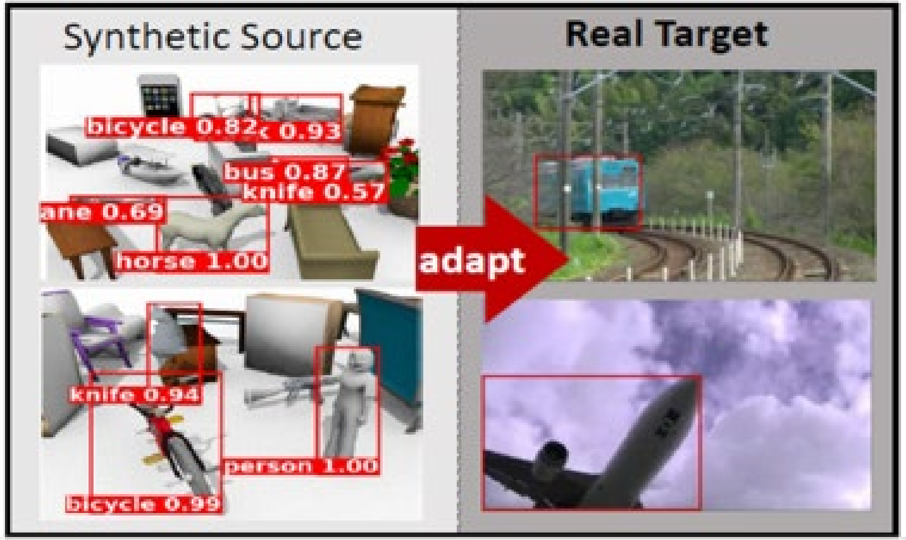
Donahue, J., et al. (2014). Decaf: A deep convolutional activation feature for generic visual recognition. In ICML.

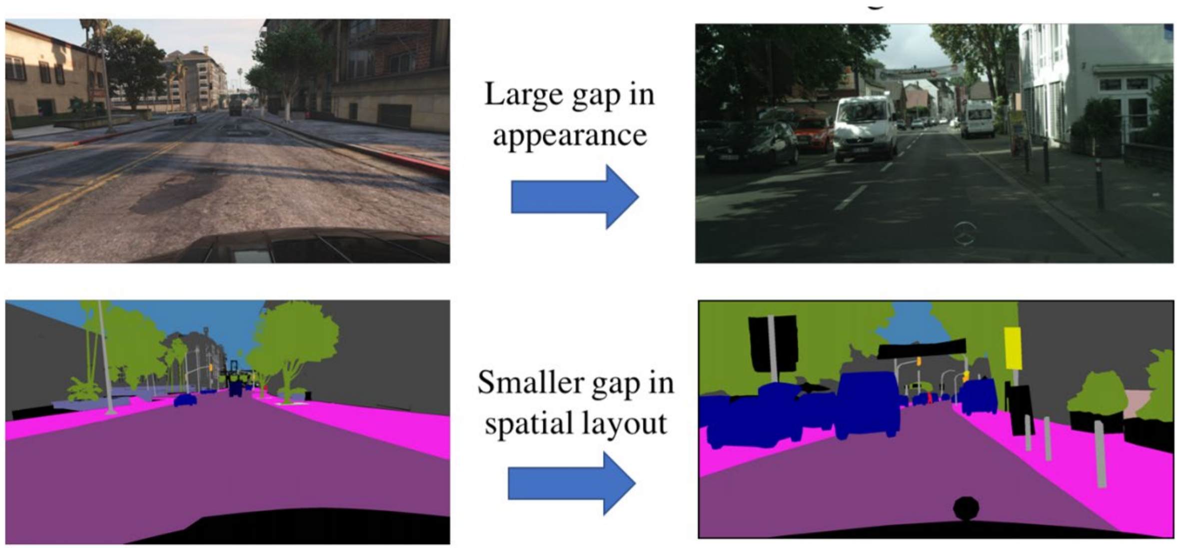
Benchmarks

DomainNet - VisDA Challenge 2019

- The Visual Domain Adaptation (VisDA) dataset is a large-scale testbed for unsupervised domain adaptation across visual domains.
- The dataset is the largest one for cross-domain object classification, with over 280K images across 12 categories in the combined training, validation and testing domains.
- The image segmentation dataset is also large-scale with over 30K images across 18 categories in the three domains.

Benchmarks

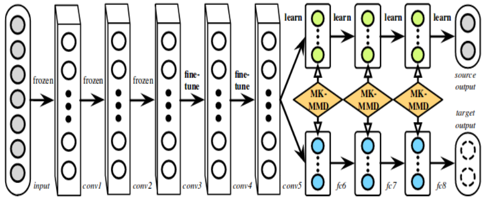




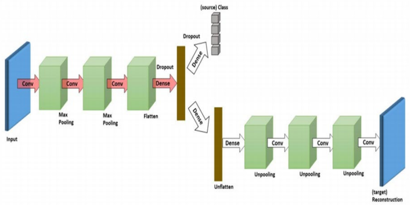
https://ai.bu.edu/visda-2018/

Tsai, Y. H. et al. “Learning to adapt structured output space for semantic segmentation”. In CVPR 2018.

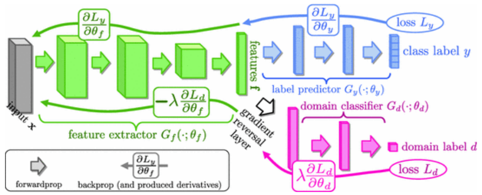
… deep DA state of the art



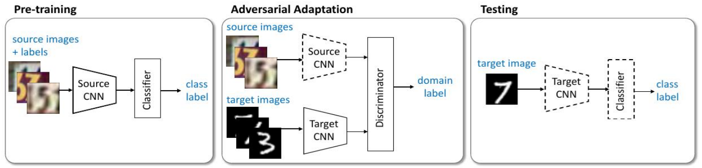
[Long et al. ICML 2015]



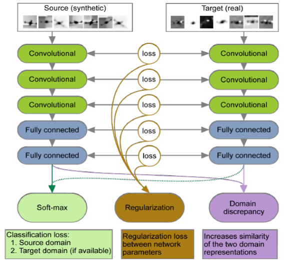
[Ghifary et al. ECCV 2016]



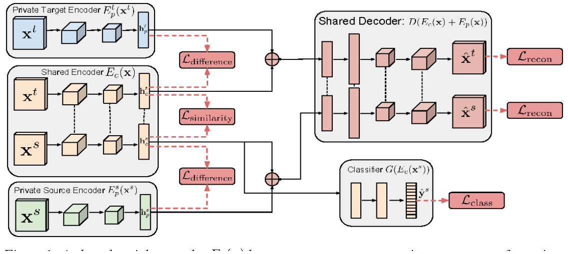
[Ganin et al. JMLR 2016]



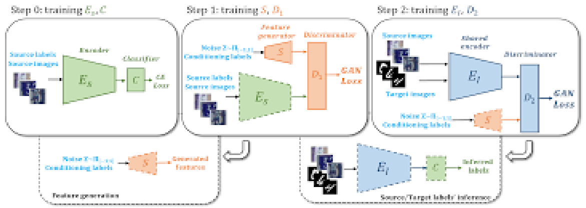
[Tzeng et al. CVPR 2017]



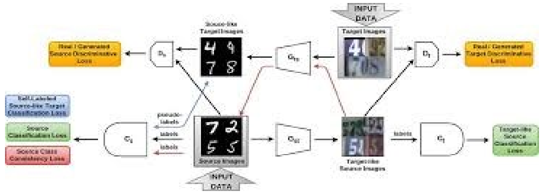
[Rozantsev et al. TPAMI 2019]



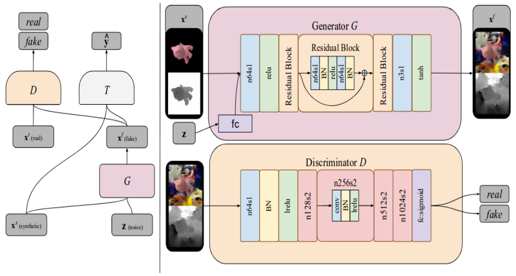
[Bousmalis et al. NIPS 2016]



[Volpi et al. CVPR 2018]



[Carlucci et al. CVPR 2018]



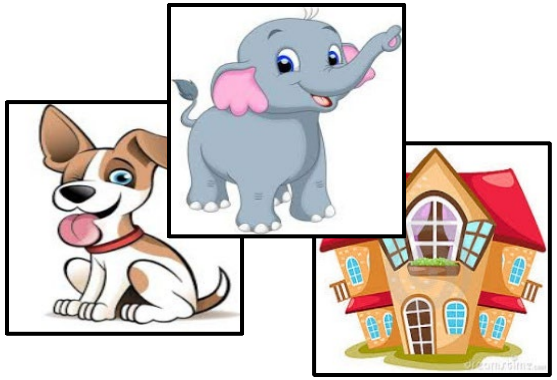
[Bousmalis et al. CVPR 2017]

Vittorio Murino

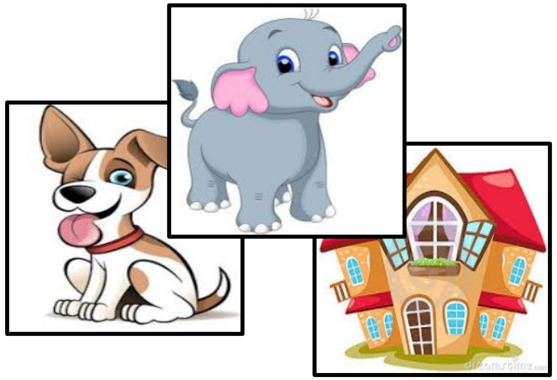
…and many, many more methods

Adversarial Domain Adaptation

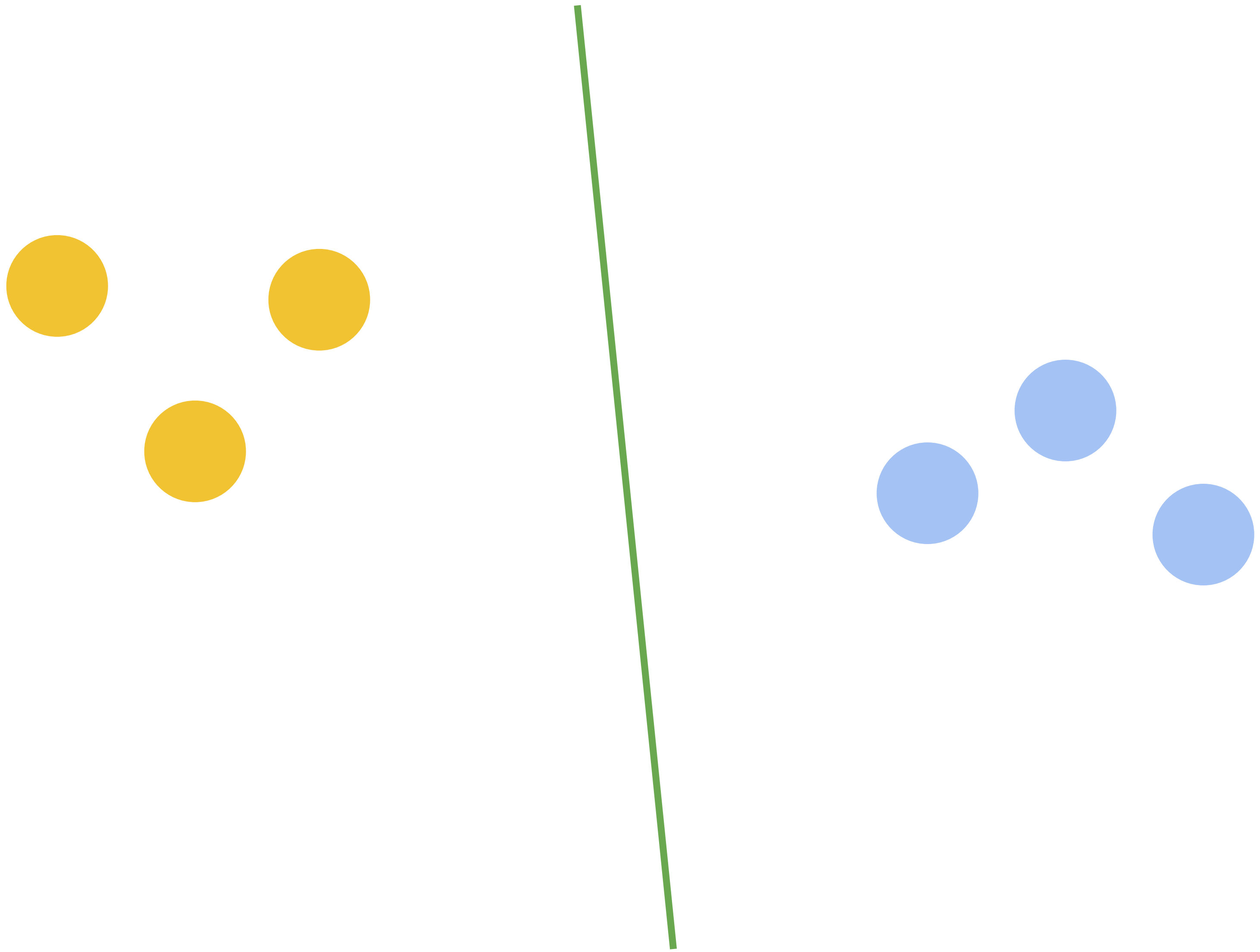
Domain Classification



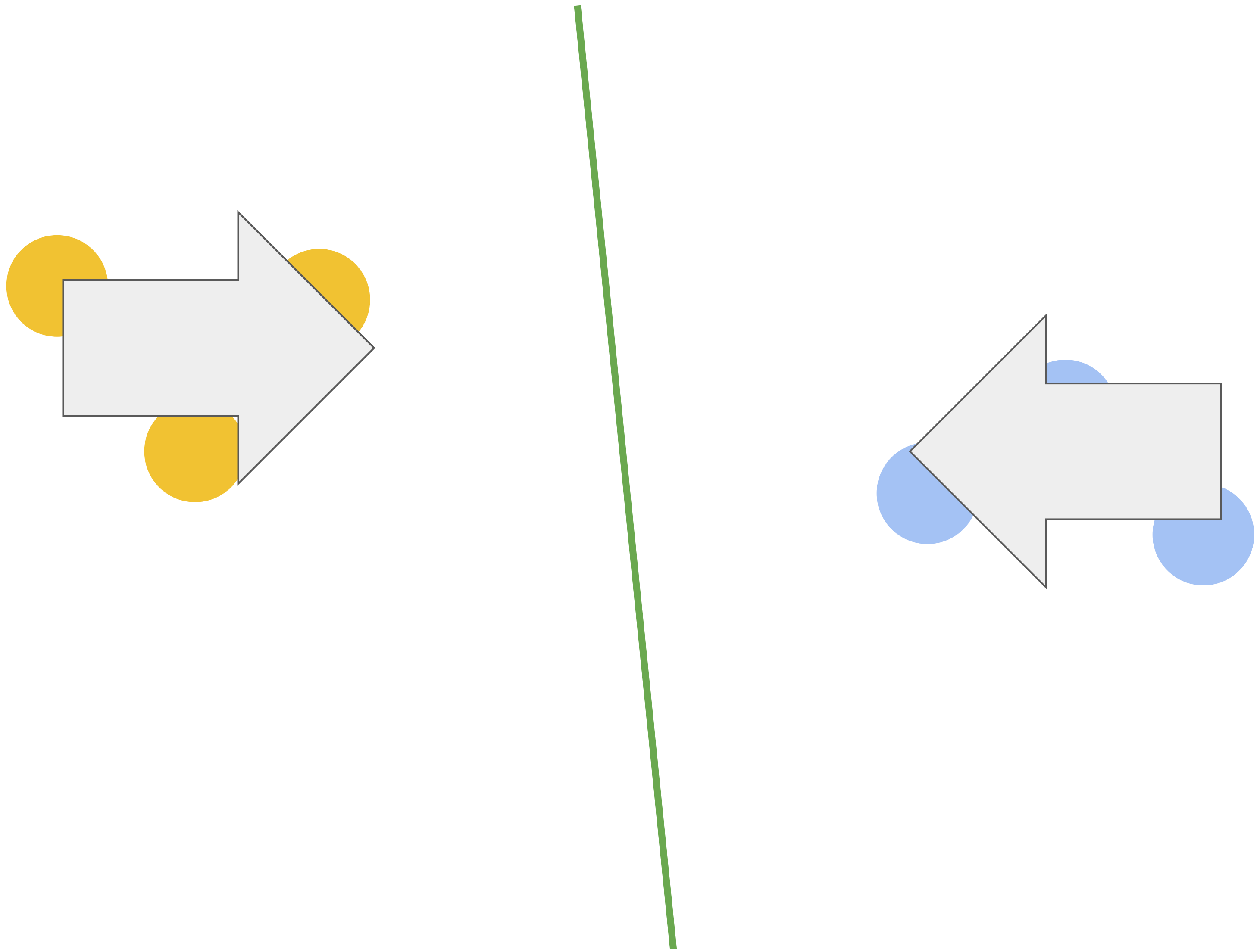
Domain Classification



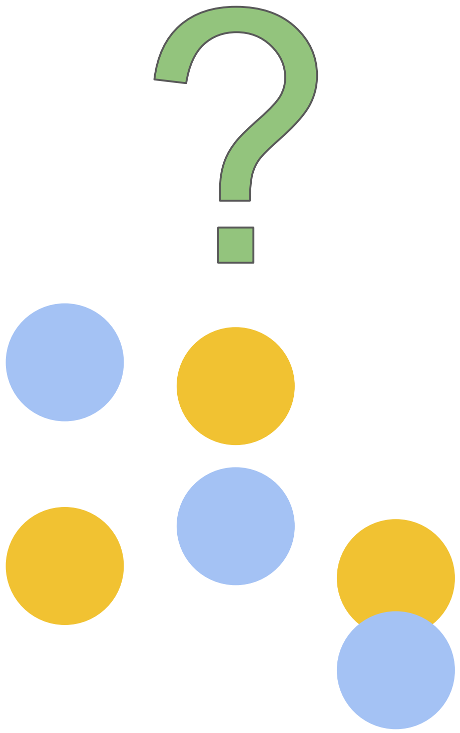
Domain Classification



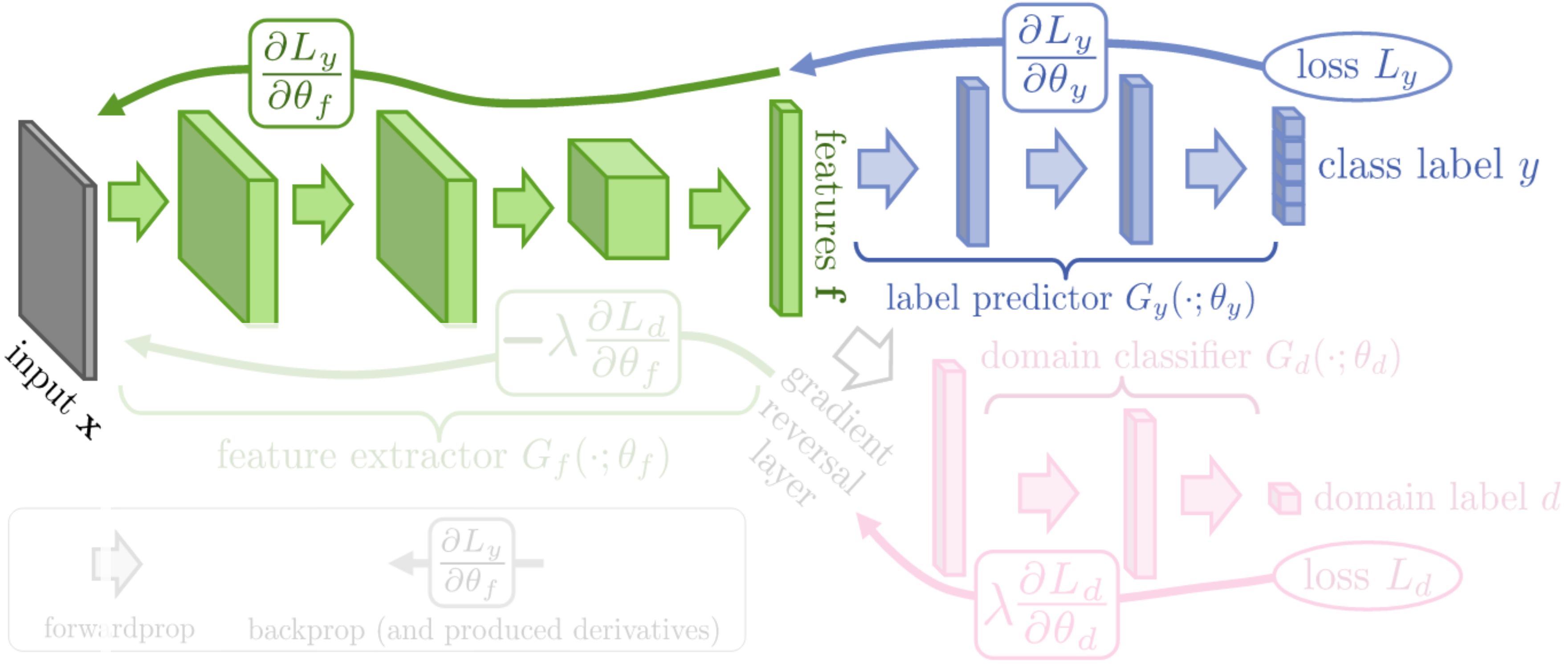
Domain Classification



Domain Classification

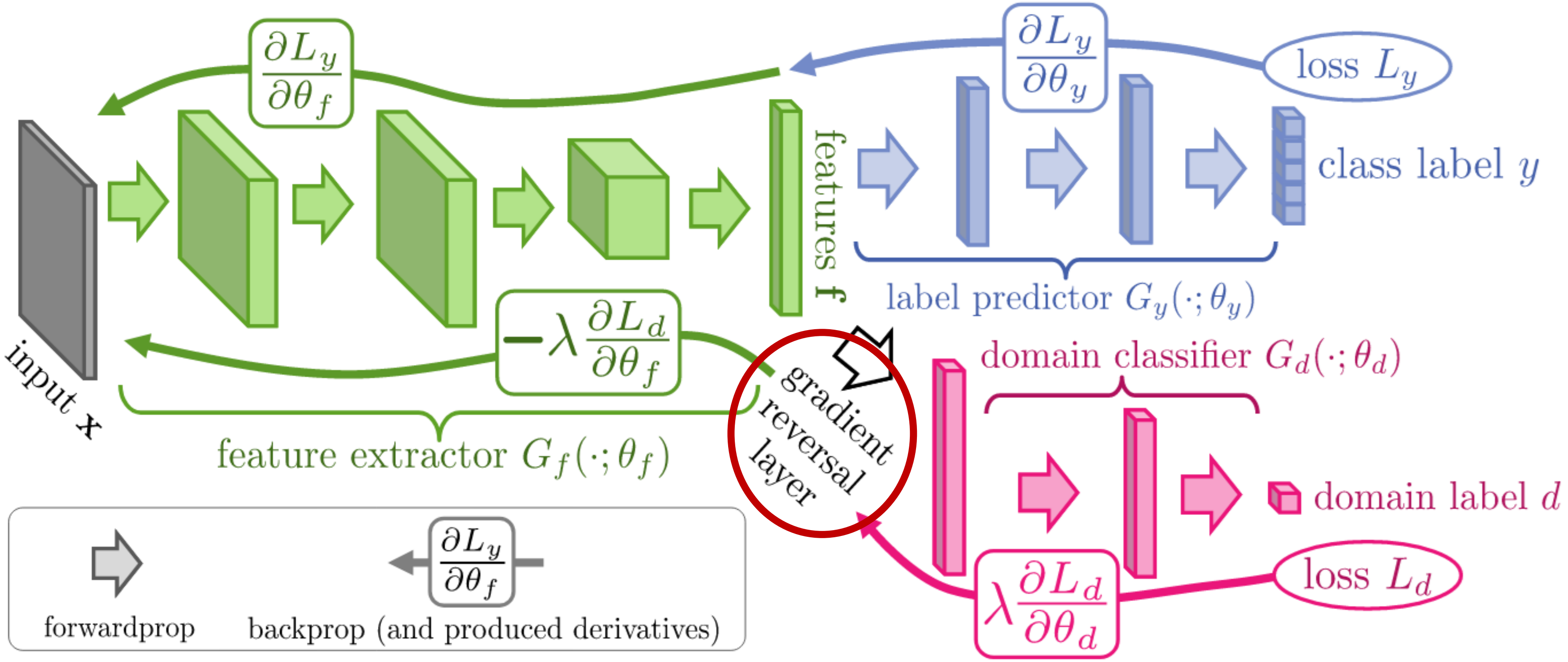


Domain Adversarial Training



Ganin, Y., et al. “Domain-adversarial training of neural networks”. The Journal of Machine Learning Research, 17(1), 2096-2030, 2016..

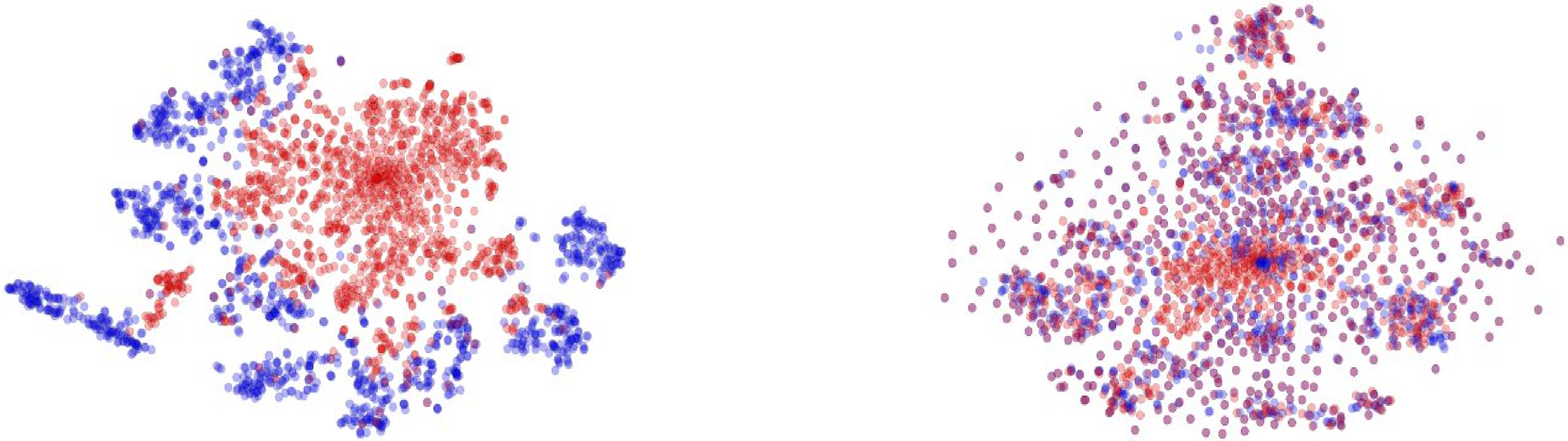
Domain Adversarial Training



Ganin, Y., et al. "Domain-adversarial training of neural networks". The Journal of Machine Learning Research, 17(1), 2096-2030, 2016.

Domain Adversarial Training

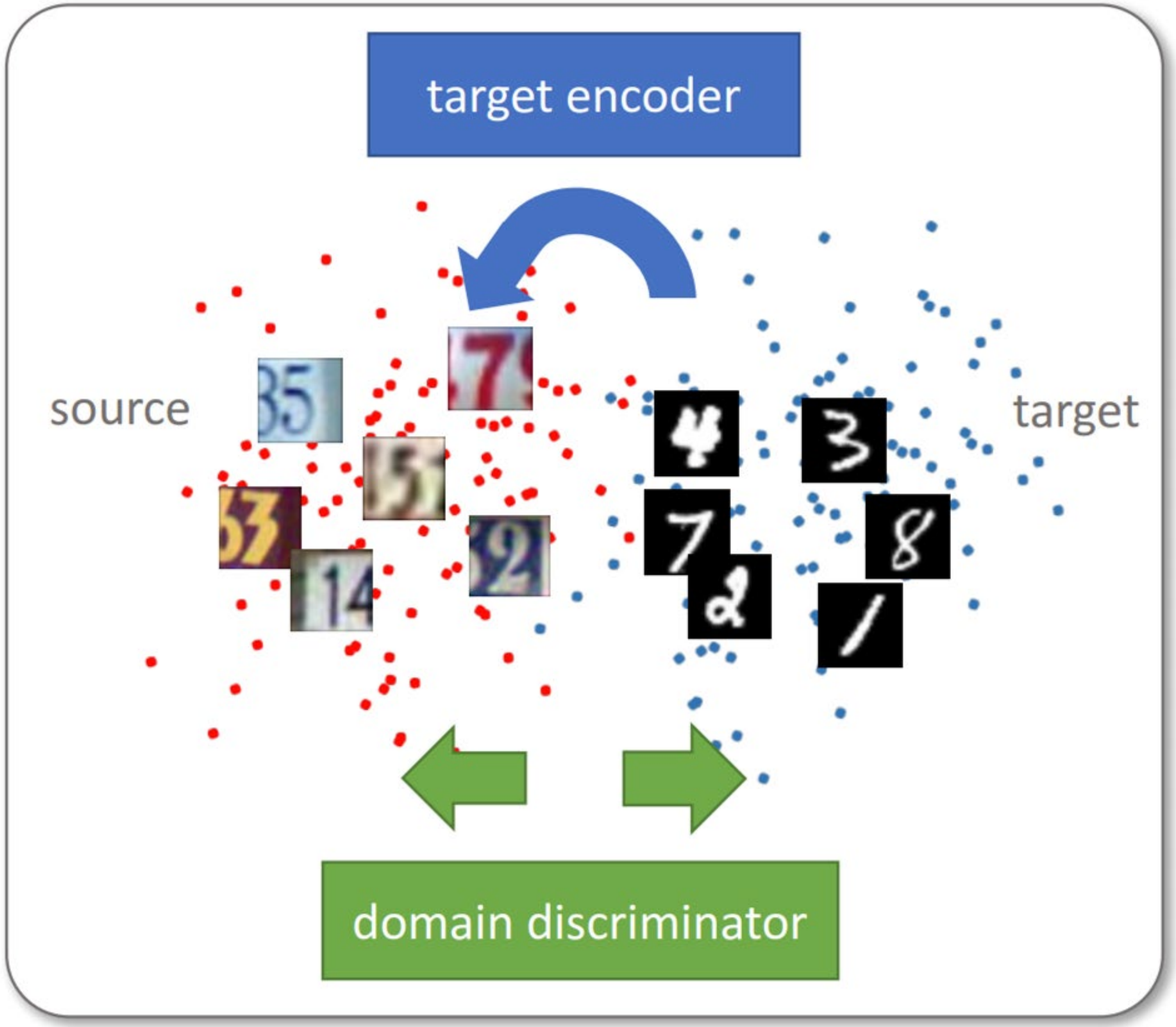
MNIST → MNIST-M: top feature extractor layer



(a) Non-adapted (b) Adapted

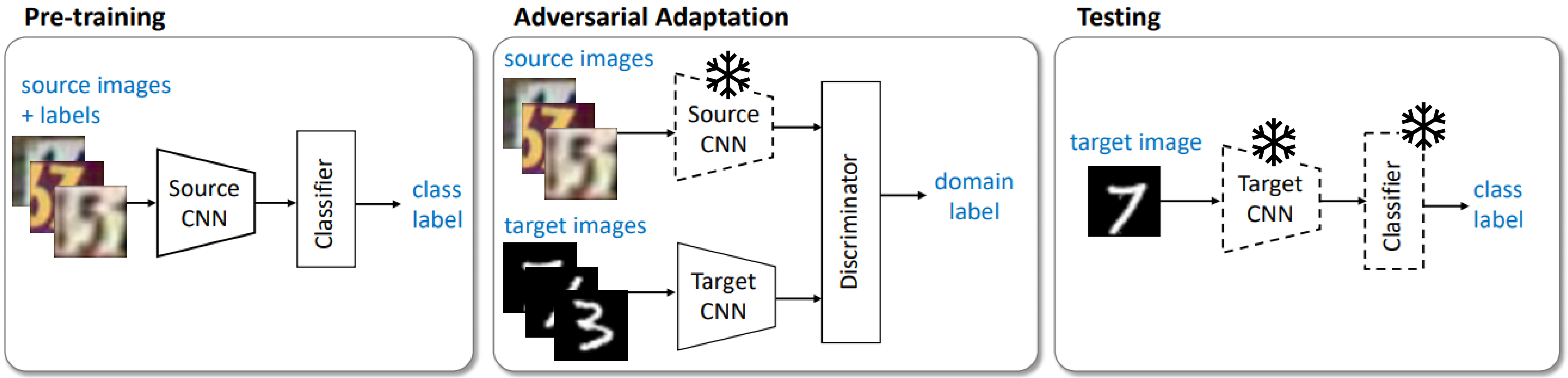
Ganin, Y., et al. “Domain-adversarial training of neural networks”. The Journal of Machine Learning Research, 17(1), 2096-2030, 2016..

Adversarial Discriminative Domain Adaptation – ADDA



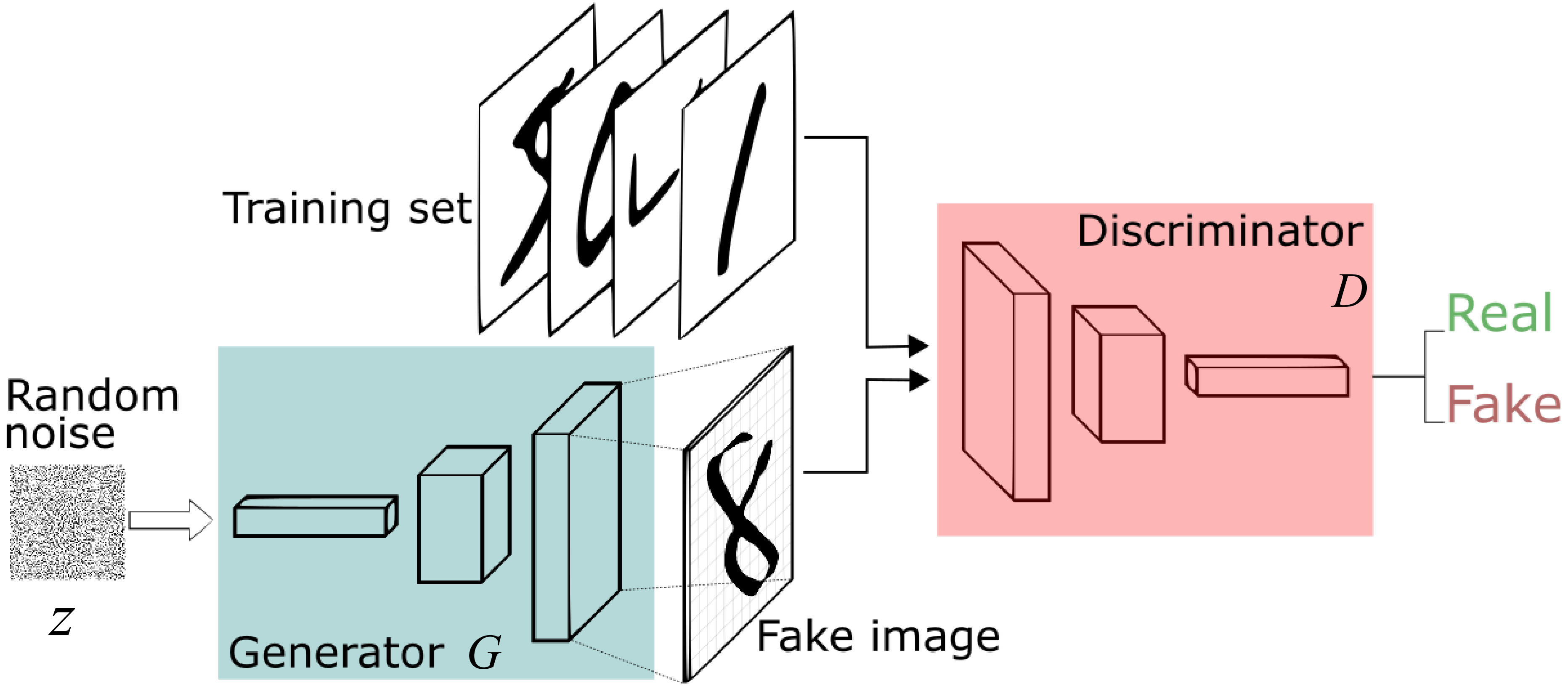
Tzeng, E., Hoffman, J., Saenko, K., & Darrell, T., ” Adversarial discriminative domain adaptation”. In CVPR 2017.

ADDA - Adversarial discriminative domain adaptation



Tzeng, E., Hoffman, J., Saenko, K., & Darrell, T., ” Adversarial discriminative domain adaptation”. In CVPR 2017.

Generative Adversarial Networks (GAN)



$$\min_G \max_D V(D,G) = \mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{z}\sim p_{\boldsymbol{z}}(\boldsymbol{z})}[\log(1 - D(G(\boldsymbol{z})))].$$

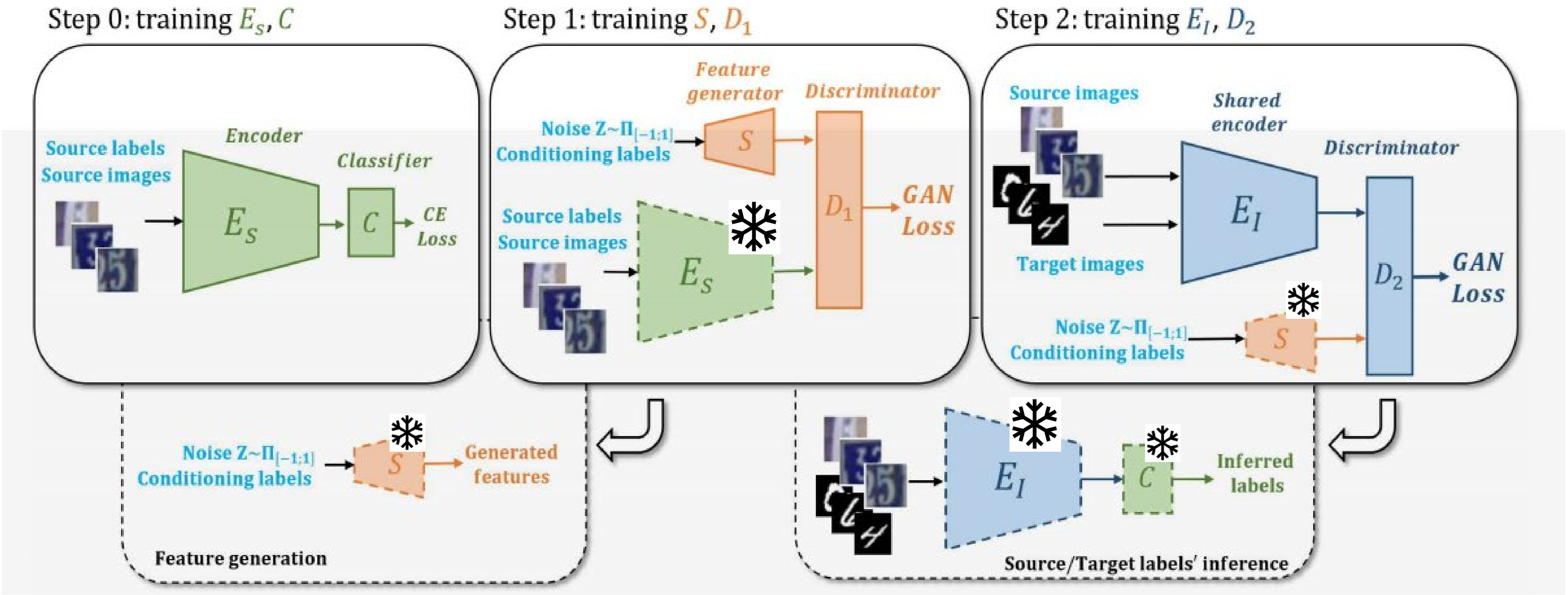
https://wiki.pathmind.com/generative-adversarial-network-gan

Domain Invariant Feature Augmentation (DIFA)

Train a deep classifier on the source domain

Adversarially train a **sampler *S*** to learn the source feature distribution (*augmented feature space*)

Adversarially train a new **shared encoder E_I** to match source and target distributions

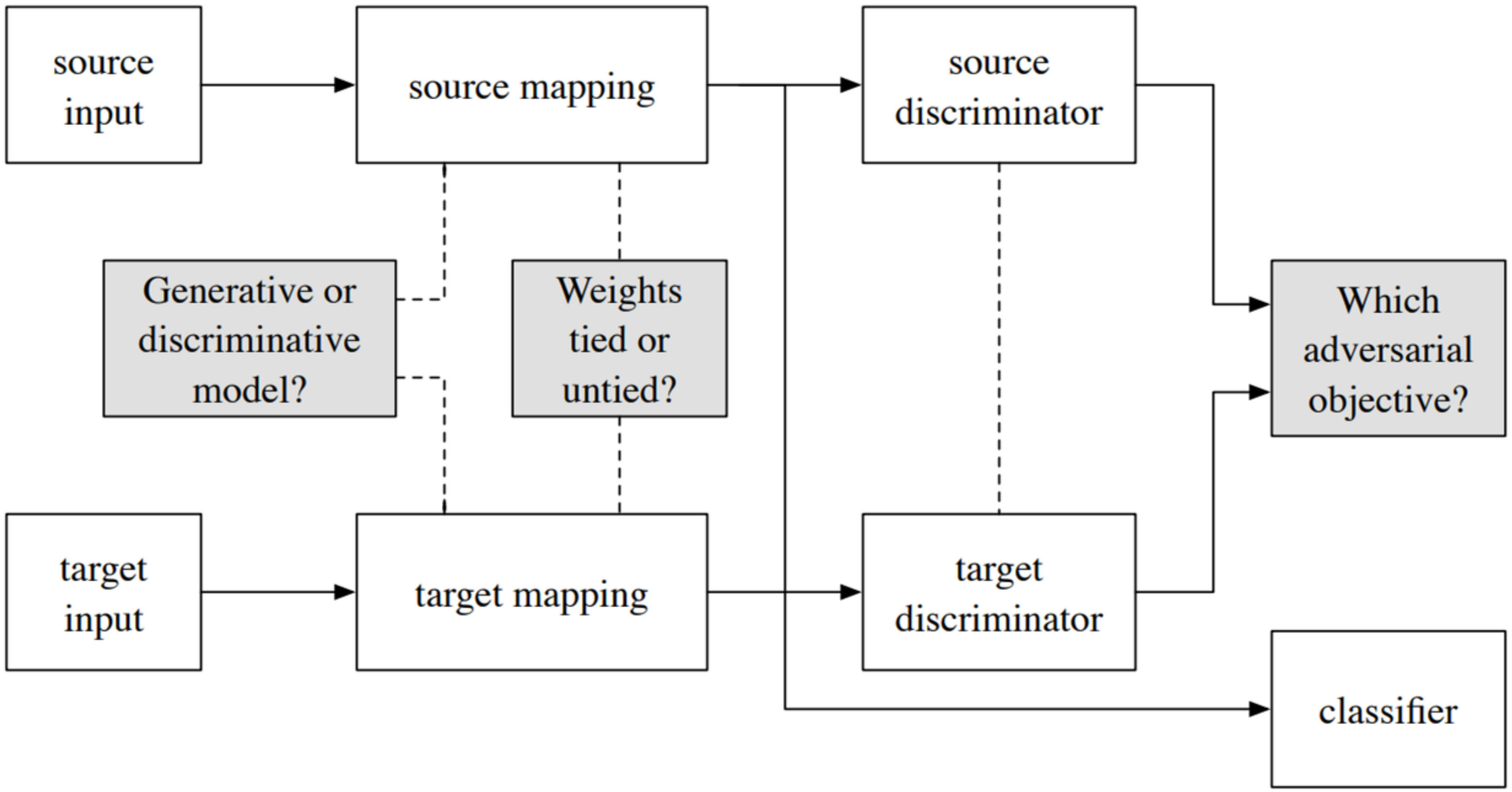


Improves ADDA by

1. Sampling source feature to perform augmentation
2. Making the encoder domain invariant (anchoring to the source)

Volpi, R., Morerio, P., Murino, V. "Adversarial Feature Augmentation for Unsupervised Domain Adaptation", CVPR 2018

In general, ADDA-like approaches …



Tzeng, E., Hoffman, J., Saenko, K., & Darrell, T., " Adversarial discriminative domain adaptation". In CVPR 2017.